

Mobile Vapour detection and identification of Synthetic drugs

PENTION

(HORIZON-CL3-2024-SSRI-01-02)

#	Organisation	Short	Type	Country
1	Mobility Ion Technologies SL (Coordinator)	MION	SME	ES
2	STAM S.r.l	STAM	SME	IT
3	SN Innovative Technologies INC	ITI	SME	Canada
4	Precision Plus Consulting Ltd	PPC	SME	UK
5	University of Sannio	USN	RTO	IT
6	Netherlands Police	NP	LEA	NL
Associated Partners (AP)				
1	RIEC Limburg	RIEC	End user	NL
2	Spanish Tax Office	AEAT	End user	ES
3	Spanish National Police	MIRPN	LEA	ES
PENTION Advisory Board (AB)				
1	Spanish Guardia Civil	GUCI	LEA	ES
2	Centre for Security Studies	KEMEA	RTO	GR
3	Smiths Detection	SMD	Industry	UK



*“In 2023, the Dutch police dismantled a record **151 drug laboratories**, a massive 44 percent more than in 2022 and the number **increased by 50 percent**, from 83 in 2022 to 125 in 2023”¹*

PENTION technology will offer an innovative ground breaking tool to police forces, and customs authorities, enhancing their capability to **detect Synthetic drugs** during the whole value chain, from manufacturing through to transportation, to the final distribution.

PENTION proposal includes Netherlands Police and the Region of Limburg (Associated Partner) where **the problem of synthetic drugs is significant and where this innovative solution would be incredibly impactful, reducing the availability of synthetic drugs, thereby saving lives.**

¹ <https://nltimes.nl/2024/05/21/record-number-drug-labs-busted-last-year-sharp-rise-heroin-synthetic-drugs>

ABSTRACT

Drug production and trafficking is a major societal challenge globally, with a high focus on the EU. The drug trade wreaks devastation throughout its value chain: from overwhelmed government administrations, to violence and brutality of the drug cartels, to citizens who succumb to addiction or fatal overdoses. On top of that, we now need to deal with a quickly evolving and very dangerous situation: the proliferation of synthetic drugs, including new psychoactive substances (NPS), which represent a real threat to society, causing serious cross-border threats to health. Synthetic drugs have already changed the traditional drug production and supply model to the extent that traditional seizing methods do not work efficiently and as effectively anymore. This situation forces the authorities to develop and implement new strategies to combat their trade which is based on the importation of chemical precursors from Eastern countries with local production, highly concentrated in different EU areas, such as the case of the Limburg Region (The Netherlands).

In addition to such new strategies, novel technologies, with better performances are required for the detection, specifically in terms of sensitivity, selectivity, versatility, portability and wider databases; this is the objective of the PENTION project which aims to develop a new vapour detector that relies on a very innovative concept of a differential mobility analyser (DMA), which classifies molecular ions based on their electrical mobility, followed by mass spectrometry analysis. This analysis generates a chemical fingerprint of the samples, and after the processing with intelligent detection routines, it becomes possible to accurately classify and detect new drugs and precursors, as well as potential manufacturing sources. The project involves the Netherlands Police and the Limburg region (associated partner) where the problem of new drug synthesis is exponentially increasing year by year.

1. EXCELLENCE # @REL-EVA-RE@

1.1 Objectives and Ambition # @PRJ-OBJ-PO@



➤ The problem and the objectives

Drug trafficking and production is one of the most profitable illegal businesses at the international level. Historically, it had a high impact in Europe with a long history of producing and consuming synthetic drugs as EMCDDA² (currently EUDA - European Union Drug Agency) has already reported from 2015³, being EUDA the reference point on drugs and drug addiction information in Europe. This illegal business not only has fatal consequences for the public finances, with an estimated black market of EUR 30 billion per year according to the EU Drug market report in 2019, and "international crackdown nets synthetic drugs worth USD 1.05 billion"⁴ according to INTERPOL, but a **major risk to citizens**. The impact of drugs on the population has a clear **direct impact** on the health of the consumers, with **more than 6,400 deaths involving one or more illicit drugs reported in 2022** in the European Union⁵, which is a 7% higher than in 2021, but an even more **indirect impact** on the families and society. Actually, the proliferation of illegal drugs is fully linked to **violence, lack of security** and other social problems such as **exploitation of minors and the emerging of criminal groups**⁶ in Europe. In this context, "identification and disruption of high-risk criminal networks" has **the highest priority of EUROPOL** within the EMPACT 2022-2025 strategy⁷.

On the one hand, many efforts have been made by European authorities to control the entire drug trade chain, from

² European Monitoring Center for Drugs and Drug Addictions, currently updated (2 July 2024) to EUDA (European Union Drug Agency)

³ PERSPECTIVES ON DRUGS Synthetic drug production in Europe

⁴ <https://www.interpol.int/News-and-Events/News/2024/International-crackdown-nets-synthetic-drugs-worth-USD-1.05-billion>

⁵ Drug-induced deaths – the current situation in Europe (European Drug Report 2024)

⁶ <https://www.europol.europa.eu/media-press/newsroom/news/europol-report-identifies-most-threatening-criminal-networks-in-eu>

⁷ <https://www.europol.europa.eu/crime-areas-and-trends/eu-policy-cycle-empact>

the drugs/precursors' entrance in Europe (or production inside Europe) to their distribution and consumption. However, this market is complex, involves tons of untraceable transactions (e.g., simply consider the invisible traffic footprint from the dark web), and is very dynamic. As a result, **drugs are still on our streets**, and according to EUROPOL⁸ “*The production of synthetic drugs is becoming more sophisticated and diverse, and there are greater opportunities for drugs to be produced nearer to consumer markets in the EU*”. This “dynamism” in the synthetic drugs trade, including NPS (New Psychoactive Substances), is fatal for the EU. The following facts still stand strong:

- **Drugs are produced in the cities with dual-use chemicals and they are easily synthesized**, as the EUROPOL reported: “*staff lacking in qualifications but properly skilled can synthesize psychoactive substances using incredibly basic instruments*”⁹,
- **Their trade time is very short (synthesis in done in hours or days¹⁰)**, with no need for international transportation of the final drugs,
- **A new clandestine lab can be built in some days within a house**,
- **The variety of drugs is very high, with new substances developed every year**. To have one idea, at the end of 2023, the EUDA was monitoring over 950 new psychoactive substances¹¹, 26 of which were first reported in Europe in 2023, and this number is growing very fast,
- Most of the precursors are **not even illegal yet. Conversely, the legal standing** (the status as legal/illegal) **substance of a chemical is crucial to managing it**. For a drug to be considered illegal for trade, it must be included in the United Nations drug conventions for this purpose, belonging to families of substances with specific chemical formulations. If one substance is not illegal, authorities do not have competence for seizure.

At this point, it is important to have an **international outlook** to prevent future problems (Figure 1), fentanyl, a NPS, is currently generating fatal consequences in the US: “*For the first time in our country's history, drug overdoses led to the deaths of over 100,000 Americans in a single year, with nearly 9 out of 10 cases attributed to synthetic opioids like fentanyl*”. Even though EU figures are not yet dramatic, the number of deaths due to fentanyl in the EU has increased by 16.4% since the last year¹² and may soon become a major problem. This data has forced the US authorities to release the **Opioid/Fentanyl Detection program**¹³ aiming at “*research novel technologies for detecting fentanyl vapour and particles*” which has a similar objective as the proposed PENTION project. Additionally, Methamphetamine is the second cause of death in the US due to synthetic drugs, after fentanyl. Currently, the main problem in the EU is focus on **synthetic drugs**, especially **amphetamine, methamphetamine and ecstasy (MDMA)** as EUROPOL is alerting in its website¹⁴. According to EUDA, “*Europe remains a significant producer of synthetic drugs*” and this data is corroborated with the following data:

- Increment around 50% in the number of synthetic drug labs identified in The Netherlands in 2023,
- 242 methamphetamine laboratories were dismantled in EU in 2022 (versus 224 in 2021),
- 48 MDMA laboratories were dismantled in EU in 2022 (versus 25 in 2021),
- The seized PMK precursor for MDMA in the EU was 19.9 tonnes in 2022 (versus 2.6 tonnes in 2021).

This remarkable data clearly shows the big systemic problem of the EU against the synthetic drugs, especially in The Netherlands and in Spain where most of the dismantled MDMA production sites in 2022 were reported¹⁵, so the **PENTION project will be focused on synthetic drugs**. However, the **results of the project (the technologies) are directly applicable to the case of NPS**, such as fentanyl and new opioids.

As stated, one problem of the synthetic drugs is the **new manufacturing and distribution approach**. In the **traditional drug market** (Figure 2), for example for cocaine and heroin, the raw materials are processed in remote geographical zones and, then transported (typically through containers or by people) as a final product to far away countries whose consumers drive demand. For example, a big part of the cocaine is imported to the EU from South America, and heroin from Afghanistan and Iran. In this case, one of the main points for customer officials to seize drugs are at **border crossing points** in seaports, land crossing points or in airports. Some solutions are available for screening and detecting them, including the technology under development in the EU-funded METEOR project¹⁶.

⁸ <https://www.europol.europa.eu/crime-areas/drug-trafficking/synthetic-drugs>

⁹ NEW PSYCHOACTIVE SUBSTANCES. A CHALLENGE TO PUBLIC HEALTH. The Spanish Early Warning System

¹⁰ THE SYNTHETIC DRUG PHENOMENON (https://www.unodc.org/res/WDR-2023/WDR23_B3_CH1_Synthetic_drugs.pdf)

¹¹ European Drug Report 2024: Highlights

¹² https://www.euda.europa.eu/publications/european-drug-report/2024/drug-induced-deaths_en

¹³ <https://www.dhs.gov/science-and-technology/opioid-fentanyl-detection>

¹⁴ <https://www.europol.europa.eu/crime-areas/drug-trafficking/synthetic-drugs>

¹⁵ Drug supply, production and precursors – the current situation in Europe (European Drug Report 2024) - EUDA

¹⁶ <https://www.meteor-project.eu/>

However, synthetic drugs' trade is quite different, most of the drugs or precursors are manufactured in **China** and, to a far lesser extent, **India** and others countries recently identified such as the case of Myanmar¹⁷. The bulk imports of NPS are often done via **mail and fast parcel services**¹⁸, in smaller amounts in comparison with traditional drugs, and additionally, and even more common nowadays, mainly the precursors (dual-use substances in some cases) are transported and the final drug is manufactured locally, in the same city or country (local markets) where they are consumed and easily marketed on the **dark web**¹⁹. For the transportation from eastern countries, some techniques are currently in practise, such as hiding the chemicals among legitimate commercial goods, mislabelling the parcels, using front companies to create the appearance of legitimacy or shipping through third-party countries²⁰, so the **only alternative for controlling the entrance is the chemical detection!**

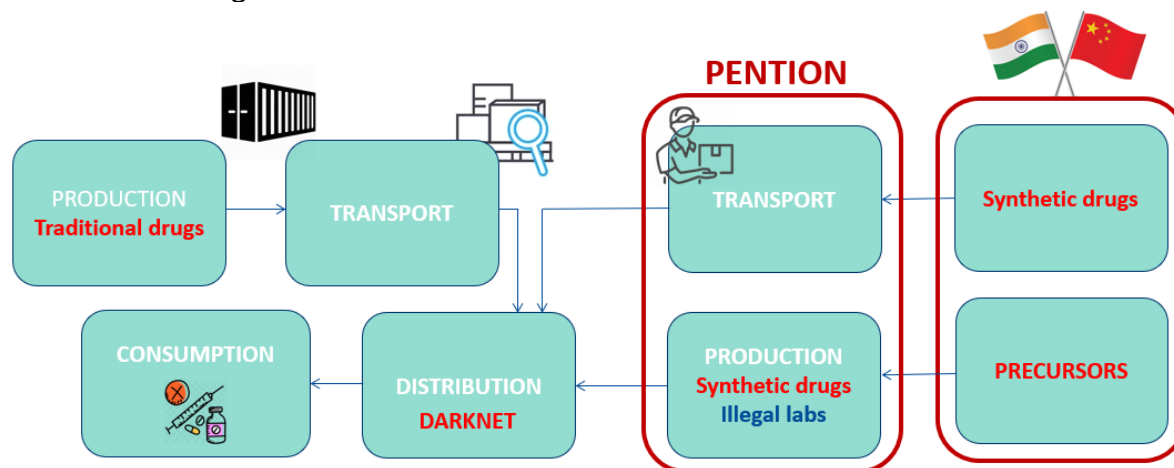


Figure 2 - Drug trade overview (traditional and Synthetic drugs, including NPS).

This proposed ground breaking new approach of intercepting the synthetic drug production will have far reaching impact on the manufacturing and distribution of synthetic drugs in Europe, where the challenge to date has been that current technologies and methodologies for the detection of these drugs are simply not capable of doing the job.

We argue **there is an urgent need** to develop and integrate **new screening processes and technologies** to detect and control these synthetic drugs, their manufacture, transport and distribution.

Authorities (Police and Customs) have already identified many challenges (PENTION KPIs have been defined according to these identified challenges), including:

Table 1 – Challenges identified by PENTION authorities

#	Challenge	Consequence	Need
1	Increase of volumes in the supply chains	More precursors available and large increase of clandestine laboratories	More accurate and sensitive technology. Automatic analysis – No interpretation by operators
2	More complex methods and imaginative concealments to import precursors ²⁰	More precursors available and large increase of clandestine labs	Technology with the capacity to chemically detect at the level of parcels or vehicles.
3	Increase and diversification of threats (due to organised crime)	Detection technologies are not easily updateable with new synthetic drug threats and their precursors	Versatile technologies based on an open database and fitted for continuous updating of the threat database by LEAs. Intelligent systems for accurate identification (AI-based)
4	Huge investments by criminal enterprise to hide the illegal activity	Existing technology solutions, even dogs, are not able to detect these clever concealments. More labs and more drugs	Highly sensitive and selective detection affordable technologies with operations 24/7 and capacity of identification. Suitable for use in court if required for chain of evidence.
5	Relative decrease of available human resources	No increase in the inspection capabilities	Technology with improved screening and detection capabilities, preferably portable.
6	Inefficiency and ineffectiveness	High FAR (False Alarm Rate)	Complementary technologies for a

¹⁷ <https://reliefweb.int/report/myanmar/interview-policing-one-worlds-biggest-drug-trafficking-corridors>

¹⁸ New Psychoactive Substances (NPS). Resource pack for informal educators and practitioners. UK government

¹⁹ <https://newsinteractives.cbc.ca/longform/the-new-frontier-of-the-drug-trade/>

²⁰ Drug Enforcement Administration - NDTA - National Drug Threat Assessment 2024

	of the current screening process based on X-ray and detector dogs alone	and low DR (Detection Rate)	multi-layered approach.
7	Possibility to make a new clandestine lab anywhere in a short period of time	More drugs available	Portable/mobile solutions for identification of new labs

In summary, LEAs, customs and other stakeholders are looking for **technology with the following performances**:

- **Efficient and fast**, which means high throughput to inspect items and minimise unnecessary disruptions to the flow of items, for example in the case of parcel/courier screening
- **Reliable (low FAR)** to avoid additional screenings with their associated costs,
- **Accurate (High DR)** to increase the detection capacity,
- **Non-intrusive** to preserve privacy,
- **Automatic results** to avoid the result interpretation from one operator which is a source of errors and avoid “external interests”,
- **User-friendly** to be widely implemented,
- **Affordable**, even though it was not an identified requirement from the end-users because they are more focused on the operational challenges, the screening cost is always important for the commercialization.
- **Explainable**, even by means of novel AI technologies, any assistive technology developed with the PENTION objectives in mind shall aid LEAs in their investigations with a data-driven interpretable explanation of any predictive, preventive, or corrective action suggested or incepted from available data.

To **meet the current needs of authorities and customs** (market need) and address the limitations of existing technologies, the PENTION project will develop a versatile vapour detector **with the sensitivity and selectivity to identify a wide range of substances, including synthetic drugs, at ultra-low concentrations**. It will be cost-effective, offering bulk detection to reduce screening costs, and provide **rapid analysis within minutes**. The system ensures non-intrusive screening, preserving the integrity of goods, and boasts a minimal false alarm rate (<5%) and high detection rate (>80/90%). Additionally, it **generates automatic results**, eliminating human error. and it will also have the capability to **easily and intelligently integrate new chemical substances into its database**, overcoming a major deficiency of current solutions. This need for advanced technology aligns perfectly with the extraordinary performance of the EVD (Explosive Vapour Detector) technology developed by MION. Therefore, the PENTION project proposes the development of a Vapour Detection system for synthetic drug detection under different scenarios and use cases.

The project will develop two versions of PENTION technology:

1. **PENTION-M**: Mobile, vehicle-mounted for detecting vapours in open areas and clandestine zones.
2. **PENTION-S**: Static, fixed setup for detection in warehouses and cargo storage places.

The PENTION project-specific **objectives** have been structured into **5 areas supported by specific project results** (Table 2) and key performance indicators (KPIs) (Table 3) addressing the complete scope (*in italics*) of the call.

Table 2 - Results of the PENTION project according to the expected objectives

Objectives	Results
Call scope: <i>Development of a mature technological solution addressing EU security policy priorities in the areas addressed by the Cluster 3 work programme.</i>	
Key area 1: Technological development	
Facilitated border checks, detection of drugs , explosives, Chemical, Biological, Radiological and Nuclear materials (CBRN), and/or other dangerous materials in customs environment	PENTION will develop a screening technology with many relevant features, one of the most relevant is to have a wide range of substances in the database with the possibility to easily integrate new ones, including complex new drugs and precursors in a mobile system.
Call scope: <i>Facilitated access to civil security market for small innovators</i>	
Key area 2: Market impact	
Despite the innovation capacity of EU SMEs, these often experience difficulties in finding their way to the public markets. These difficulties are	PENTION involves four SMEs with international relevance, however, the main innovation is developed by the two EU based SMEs, supported by a RTO and a LEA,

exacerbated in markets that show restrictions of different kinds, as in the case of security.	which will help showcase the system in a relevant environment, reducing the difficulties faced by SME developers in penetrating the security market. Additionally, the PENTION AP and AB will support on the market impact since they are end-users and future early adopters.
Call scope: Improved cooperation between public buyers and small supply market actors for a swifter uptake of innovation in response to short to mid-term needs	
Key area 3: Collaboration between SMEs and security first-adopter	
The proposal should aim at fostering collaboration between SMEs from different Member States and Associated Countries; involving security end-users in the role of validator and potential first-adopter of the proposed innovations.	PENTION is based on the collaboration between technological SMEs from different member states (Spain and Italy), but also supported from SMEs in the UK and Canada and a LEA end-user in The Netherlands, which will be the first adopter of the developed solution. The technical expertise coming from Canada (ITI) is quite relevant. Some LEAs are included as AP and members of the AB.
Detection of synthetic drugs, including NPS, is a short/urgent need for the EU, if we don't want to have the figures of the US. The US launched the Opioid/Fentanyl Detection program in 2021, when the number of people death was more than 100.000 per year, too late!	
Call scope: Stronger partnerships between small and medium EU security industry and technology actors to ensure the sustainability of the EU innovation capacity in the civil security domain and reduce technological dependencies from non-EU suppliers in critical security areas.	
Key area 4: Collaboration with research centres to strengthen innovation capacity	
Fostering collaboration schemes between small companies and research and technology organisations aimed at fostering innovative technology transfer that strengthens the innovation capacity of EU SMEs/start-ups in the domain of civil security.	PENTION involves The University of Sannio in Italy which will support the SMEs in the most technological part of the project and PENTION also includes Smiths Detection (AB - big Industry) with the aim of supporting us for the future product development and its market introduction.
Call scope: Creating innovative business models that facilitate access to the market of EU SMEs and start-ups in the domain of civil security.	
Key area 5: Exploitation + Commercialisation and business plan	
MION, STAM and PPC will develop individual exploitation plans and a joint business plan for the market-ready solution. PPC will develop the PENTION business plan and the strategy to introduce the solution into the market, initially in the EU market.	Creation of a business plan for the consortium and for individual partners that are looking to exploit the project activities, mainly commercialisation of the PENTION portable system. The possibility of reaching agreements with big security players globally will be also contemplated in the commercialization plan.

In addition to the objectives related to the HORIZON-CL3-SSRI-01-02 call, the following **KPIs (Key Performance Indicators)** are defined to monitor and assess the proper evolution of the project and the fulfilment of the expected objectives. These KPIs have been defined trying **to deal with the authority challenges** defined in Table 1:

Table 3 – PENTION KPIs

KPI#	PENTION Key Performance Indicators (KPIs)	Associated challenge	M24
1a	Sampling Time / courier use case	1, 5	< 1 min
1b	Sampling Time / Vehicle + open area use case	1, 5	< 3 min
2	Analysis time	1, 5	<1 min
3	Number of substances in the database (narcotics + subproducts + precursors)	3	> 6
4	Duration of the batteries in the sampler	1, 5	> 5 hours
5	Detection Limit (key for detection at bulk level)	3, 4, 6	ppq range
6	Detection Rate (DR) for open areas use case (clandestine lab)	3, 4, 6	> 80%

7	Detection Rate (DR) for semi-close areas (car screening/courier)	3, 4, 6	> 90%
8	False Alarm Rate (FAR)	3, 4, 6	< 5%
9	Automatic Results (Level of human interpretation)	3, 4, 6	0%
10	Days of Real Testing in an Operational Environment	NA	> 2 weeks
11	Prototype TRL (Technology Readiness Level)	NA	7
12	Number of scanned items in the project	NA	> 35
13	Screening capacity for open areas	7	>100 m ² /hour
14	Dissemination: The audience reached by our mainstream articles and press releases	NA	3k
15	Dissemination: Attendance of customs/security related events	NA	4
16	Communication: # of media practitioners following the project results	NA	150
17	Collaboration: # of seminars organised with related EC projects	NA	4

➤ Current detection technologies

Depending on the type of scenario, different technologies and methodologies are already in use. For the PENTION project, three use cases have been identified (see use cases section), but the available technologies are the following:

- **Intelligence / Risk management.** This is mainly based on the intelligent analysis of the available information associated to the suspicious item (parcel in a warehouse, cargo in a container, or house in a residential area) and determine a probability. If the probability is higher than a threshold, a deeper analysis is required in order to confirm the presence of the suspicion, at this level some technologies can be implemented, even PENTION.
- **Imaging scanning technology (X-rays)** is the widest implemented technology for **items (parcel, baggage, etc) screening**. X-ray technology is versatile and provides confidence by revealing what is inside items (intrusive) but has limitations in precise material separation simply due to the nature of transmission X-ray imaging concept. In addition, the detection relies on human interpretation (not automatic) which is prone to errors, even though it is typically supported by advanced visual identification algorithms, so relying solely on X-ray transmission can lead to unsatisfactory **False Alarm Rates (FAR)** and **Detection Rates (DR)**²¹. This technology is broadly implemented, where it has been developed mainly for explosives, but the **detection of synthetic drugs or precursors by imaging systems is not highly developed yet**. For example, some attempts are under development using other imaging technologies such as Terahertz imaging²². Another challenge for the imaging systems is the innovation under the smuggling techniques including bulk mixtures, impregnation of textiles and other merchandise, and form casting, have been devised to evade imaging detection which makes even harder the detection. This is due to the challenge X-ray scans face in distinguishing materials with similar densities.
- **Explosive Detection Dogs (EDDs)** are valuable assets in the detection of smuggled goods such as drugs, tobacco, and even currency transported in luggage or in suspicious items, they can be also used for detection of illegal/ clandestine manufacturing laboratories. This approach has demonstrated global success for several decades, but it has important drawbacks. **Firstly**, screening throughput of EDDs is quite limited, however, they have the advantage to screen in motion, which is very useful for some types of scenarios, like the detection of clandestine labs or in dynamic scenarios, like the courier warehouses, however, smugglers have resorted to using chemicals designed to confuse and disrupt dogs' sniffing abilities. **Secondly**, these highly trained canines can experience reduced detection performance due to fatigue, but **they may not manifest in external symptoms**, requiring rest periods after their working cycles, typically lasting between 20 to 30 minutes. **Thirdly**, their sense of smell can be easily saturated and is influenced by external factors such as temperature and humidity. **Lastly**, the operation of sniffer dogs is very costly. TSA (Transport Security Agency in the US) declares the following: *"TSA pays \$218,000 in startup costs **per dog** during training, then \$158,000 per year after that, says Greg Soule, a TSA spokesperson. The money covers the salary of the handler, training, certification, veterinary services, kennelling and dog food"*²³. Which means a total cost of 1.48 M€ per dog during the 8 years of active work, and this cost is considerably higher than the cost of the one unit of PENTION technology (acquisition cost around

²¹ No official data of FAR and DR are available. This information is considered as classified, however the FAR for X-rays to container screening is around 66% according with data obtained from the Port of Rotterdam. 2/3 of the containers that provides any suspicious in the X-Ray analysis don't contains any illegal item inside.

²² K. Kawase, Y. Ogawa, Y. Watanabe, and H. Inoue, "Non-destructive terahertz imaging of illicit drugs using spectral fingerprints," Opt. Express 11, 2549-2554 (2003).

²³ <https://science.howstuffworks.com/bomb-sniffing-dog.htm>

650,000 € + 25,000€ annual maintenance) that has the capacity to work 24/7 for more than 10 years. In addition to this, sustaining continuous operation necessitates maintaining a group of **several dogs**. **Additionally**, EEDs can be trained for a limited list of substances and the training for detecting new threats requires some time which is not suitable for the case of synthetic drugs and NPSs.

- **Explosive Trace Detection (ETD)** which are mainly used as a secondary (confirmatory) technology, this is typically used for detection of explosives, and some drugs. ETDs operate by physically collecting traces at the particle level (in nanograms) using a paper swab. **The main limitation of ETDs is the low selectivity** of the technology behind, which is typically an IMS (Ion Mobility Spectrometers), which limits the sensitivity. In addition, the need to physically collect the particles limits the capabilities. However, ETDs are portable and the inclusion of new substances is not a relevant challenge since the detection is based on the physical particle collection, versus the vapour detection which depends on the vapour pressure of the target substances.

The Table 4 summarizes the performances of the current detection systems vs. the proposed PENTION technology.

Table 4 – PENTION technology versus existing solutions

	X-Rays	EDDs	ETDs	PENTION - Vapour
DR	Low (<20%)	Medium (no data)	Low (no data)	>80%/90%
FAR	High²⁴ (>50%)	Medium (no data)	Low (no data)	<5%
Portable	No	Yes	Yes	The sampling part
Valid for Synthetic drugs?	No	Partially	No yet	Yes
Valid for clandestine lab identification?	No	Partially	No	Yes
Intrusive	Yes	No	No	No
Screening costs	Low	High	Medium	Low
Automatic Results	No	Yes	Yes	Yes
Easy use	Yes	No	No	Yes
		PENTION system will comply with all the identified challenges by authorities		

➤ PENTION proposal and operation

In order to comply with the current needs of the authorities and customs and to solve the limitations of the existing technologies, **PENTION project will develop a vapour detector that offers the sensitivity and selectivity required to identify at ultra-low of vapour concentrations a wide variety of chemical products with the final objective to detect synthetic drugs**, and with the capacity to easily and intelligently integrate new chemical substances in the database, which is one of the main deficiencies of the existing solutions.

MION EVD detection technology has already reported limits of detection in the order of 0.01 ppq (parts per quadrillion) with FAR lower than 1% for explosive detection²⁵ in a real environment, providing a solution with the capacity to detect very low vapour pressure substances, such as cocaine hydrochloric, in large volumes, in fact, it is the lowest limit of detection ever reported and in consequence, the only technology so far with the capacity to detect some of the most “trendy” synthetic drugs. The vapour pressure is a measure of the tendency of a material to change into the gaseous or vapour state, the higher vapour pressure, the more presence of molecules in gas phase and the easier the detection in gas phase (theoretically). In Figure 3, it is represented the vapour pressure of the target substances for the PENTION project, including the cocaine and the TNT (explosive) used as reference. MION’s DMA-MS (Differential Mobility Analyser – Mass Spectrometry) technology has been successfully used to detect a few grams of TNT and some kilograms of cocaine in containers (large volume), so the potential detection of the three

²⁴ According to Dutch Customs, in the port of Rotterdam, around 66% of the containers with an anomaly in the X-ray scanner don’t have any threat inside.

²⁵ “Reaching a Vapour Sensitivity of 0.01 Parts Per Quadrillion in the Screening of Large Volume Freight”. D. Zamora, M. Amo-Gonzalez, M. Lanza, G. Fernández de la Mora, and J. Fernández de la Mora. Analytical Chemistry 2018 90 (4), 2468-2474

new compounds is theoretically feasible, but nobody has done it before.

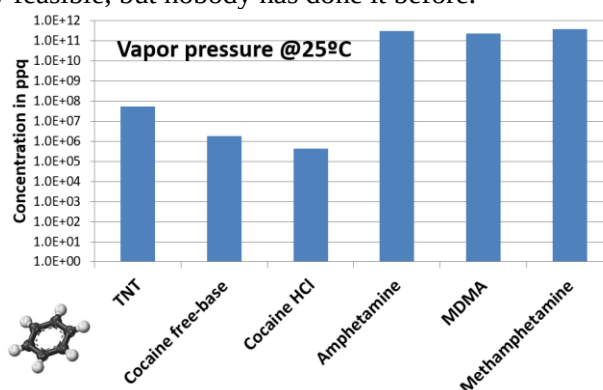


Figure 3 – Vapour pressure comparative. TNT and cocaine are used as reference. Graph elaborated by MION from several sources

The extrapolation of the PENTION results to synthetic drugs, and especially the NPS, is not direct because there are not solid studies about the vapour pressure of some of these substances. For example, in the case of Fentanyl (which has been deeply studied in the US development program since 2021), the Ewing group in the US reported a vapour pressure of 0.018 ppq for fentanyl hydrochloride.²⁶ However, the group of Gupta reported vapour pressures in the range of 5.8 ppt²⁷ and the Meng group estimated the vapour pressure at the level of 32 ppt.²⁸ This variety and discrepancy of figures is mainly due to **the lack of ultra-sensitive instruments that can confirm the results**. So, a deeper analysis using MION's technology is needed (T2.2). Some attempts to detect “fentanyl materials” have been already reported at lab level²⁹, however this is very far from a real-world application.

In the case of synthetic drugs, it is not only the detection of the final product that is relevant, but also, and in some cases more importantly, the **detection of the precursors**, or even the subproducts. The case of the precursors is complex since the manufacturing processes are evolving continuously as EUROPOL is alerting. For example, for the methamphetamine, the production process was traditionally based on the “ephedrine method”, which is based on the ephedrine + pseudoephedrine, however, an alternative manufacturing process is based on BMK³⁰ which is a common precursor also for the amphetamine, so the **capacity to easily introduce new substances in the database is critical**.

The proposed PENTION vapour detection technology is based on the concept of EVD developed by MION, named VPSCAN (<https://vpscan-evd.com/>) which is based on the combination of a high flowrate sampling unit (sampler) and the DMA-MS analysis as it will be presented later on. As discussed, the project will develop **two versions of the PENTION technology**, **one Mobile (PENTION-M)** which will be vehicle-mounted and will have the capacity to be transported in a van for the detection of tiny concentrations of target vapour in open areas and the possibility to detect clandestine manufacturing zones and **one Static (PENTION-S)** which will be installed in a fixed setup and valid for the detection in scenarios like warehouses, cargo storage places, etc. Both versions will be based on the same technology, but with different sampling systems.

Table 5 – Two versions of PENTION technology to be developed in the project

	PENTION-M	PENTION-S
Applications	Clandestine labs, events, forensic, etc	Screening processes
Target substances	Final products (main) Sub-products	Precursors (main) Final products
Sampling system	Portable wireless	Fix installation
Analysis system*	Embarked	Fix installation
Challenges	High sensitivity in a portable system	High sensitivity in a rapid screening system

*It is important to note that only one analyser based on DMA-MS will be developed and built in the project, it can operate embarked in the vehicle or in a fix installation.

PENTION-M version is based on the integration of the DMA-MS technology into a mobile system, typically a van,

²⁶ Ewing, Robert G., Nims, Megan K., Morrison, Kelsey A., Hart, Garret L., Avalos, Nancy M., and Denis, Elizabeth H. vapour detection and vapour pressure measurements of fentanyl and fentanyl hydrochloride salt at ambient temperatures. United States: N. p., 2022. Web. doi:10.1039/d2an01149b.

²⁷ Gupta, P. K., Ganesan, K., Gutch, P. K., Manral, L., & Dubey, D. K. (2008). vapour pressure and enthalpy of vapourization of fentanyl. Journal of Chemical & Engineering Data, 53(3), 841-845. <https://doi.org/10.1021/jc7005067>

²⁸ Y. Meng, A. H. Lichtman, D. T. Bridgen and B. R. Martin, NIDA Res. Monogr., 1997, 173, 201–224.

²⁹ Charles D. Smith et al. Detection of N-phenylpropanamide vapour from fentanyl materials by secondary electrospray ionization-ion mobility spectrometry (SESI-IMS), Talanta Open, Volume 5, 2022, 100114, ISSN 2666-8319, <https://doi.org/10.1016/j.talo.2022.100114>.

³⁰ benzyl methyl ketone, or phenyl-2-propanone, ‘P-2-P’

and has the capability to collect air samples using the portable sampler in different areas, typically rural or industrial, where there is a suspicion of clandestine synthetic drugs labs, see Figure 4.

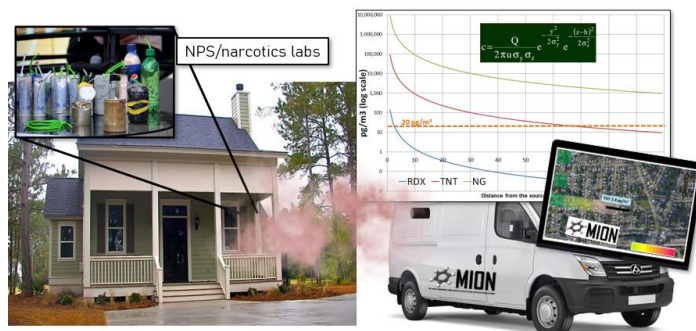


Figure 4 – PENTION-M concept

EVD technology has the capacity to detect, identify and quantify vapour concentrations in the order of a few femtograms/m³ in real-world environments. This sensitivity allows the detection of substances in moderate quantities (<1Kg) at distances between 50 and 100 m. According to the vapour pressure graph (Figure 3), the detection capacity for synthetic substances such as MDMA, amphetamine and methamphetamine, is even higher, as the vapour pressure is around one million times higher; this is of great interest as it makes it possible to detect clandestine labs by the vapour emissions at distances close to one Km.



Figure 5 – PENTION grid approaching strategy

The final identification of the lab locations will be done by an intelligent analysis performed by USN by correlating the available data in a grid approaching strategy. Figure 5 shows an example of the vapour concentrations of MDMA in an urban area. Vapour diffusion in an open area follows dispersion models³¹ with influence of many factors, but mainly weather conditions. This study will be done by USN and implemented in the detection routines.

PENTION-S version is a solution for the detection of synthetic drugs in static scenarios, typical for the PENTION project is the detection of substances in parcels within the courier warehouse facilities. For this operation a static sampling system with automatic analysis of the samples is proposed. For PENTION project, detection at the level of the closed box is planned with the aim of being the first detection technology with the capacity of detecting a synthetic drug still residing in a box by vapour detection.

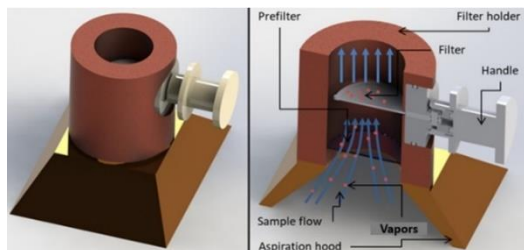


Figure 6 – PENTION-S Concept.

Figure 6 represents the concept of PENTION-S which is a fixed sampling unit collecting vapours from the target parcel/box/item. Vapours are collected in the filter and then analysed in the analyser. One objective of the PENTION-S system is the characterization of the vapour emissions of synthetic drugs inside parcels with different levels/layers of concealment, representing the real use case.

The PENTION concept has sparked **great interest in the customs and law enforcement community**, therefore, the **PENTION Advisory Board (AB)** has been created at the proposal stage as an alternative to being partners due to the budget limitations for non-SME partners in the SSRI-01-02 call. Associated partners are also part of the AB, currently, the PENTION AB consists of Limburg Region (RIEC), Spanish Customs, Spanish National Police, Spanish Guardia Civil, KEMEA and Smiths detection; however, it is planned to include more members during the initial stages of the project. A specific task (T1.5) has been defined for the AB management due to the high interest in the PENTION concept in order to optimize the technical developments and maximize the impacts of the project.

Within this framework, the prioritisation of implementing new screening technologies stands as a crucial objective for customs and police authorities. **The primary focus of the PENTION project is the development and validation of a technology based on vapour detection for synthetic drugs.** This innovative approach involves collecting a representative air sample from around the item, followed by a chemical analysis of its contents. These technologies offer a non-destructive and minimally intrusive approach, thereby enhancing safety for screeners. Moreover, the fully automated nature of these methods eliminates the need for human interpretation, contributing to increased efficiency and reliability. The key performance aspects can be summarised as follows:

³¹ https://en.wikipedia.org/wiki/Atmospheric_dispersion_modeling

- **Wide Range of Substance Detection:** Capable of detecting several substances, not limited to the drugs, but also, precursors, chemicals, degradation products, etc in an open database mode where the **inclusion of a new substance** can easily be done by the inclusion of the substances parameters once they have been identified in the lab (no more than 1 week).
- **Versatile Application:** Suitable for diverse scenarios due to the separation of the sampling and the analysis processes, with a particular initial emphasis on detecting clandestine operations and trade, but it can apply to many scenarios from a group of items/parcels to a complete suspicious truck/container.
- **Cost-Effective:** Available at an exceptionally affordable cost due to the capacity of “bulk” detection. Existing technologies are based on individual analysis, the technology developed in PENTION project will have the capacity to detect at the level of truck, pallet, etc which significantly reduce the screening costs, allowing to focus the detection. In the case of the detection in open air scenarios, the possibility to have some signals from the exterior without the need for individual screening, it is a great advantage.
- **Rapid Analysis:** Yields results in approximately 1-3 minute for sampling (depending on the scenario) + <1 minutes for analysis, ensuring swift and efficient screening.
- **Non-Intrusive Screening:** Detects substances within complex and heterogeneous loads without the need to “see”, unpack or open the container, parcel, pallet, etc, preserving integrity and privacy.
- **Minimal False Alarm Rate (FAR) (<5%):** Demonstrates a very low FAR, reducing unnecessary disruptions and extra costs which will require additional selectivity in the vapour collection process. This extremely low FAR will reduce the total costs of the operation. It is expected even a lower FAR in the future.
- **High Detection Rate (DR) (>80/90%):** Exhibits a high DR, contingent on the type of substance being screened and its packing and concealment conditions. This high DR will significantly increase the seizure rate which is the final objective of the customs and finally, the desired objectives of the citizens. 90% for close scenarios / 80% for semi close scenarios.
- **Automatic Results:** Generates automatic results, eliminating the need for human interpretation which is a source of errors and consequently, increasing FAR while reducing DR.

To increase the detection efficiency, the **combination of PENTION technology with a Risk Management System is the perfect solution**. PENTION project is mainly based on the development of the hardware and the detection prototypes, but it includes some intelligent based on atmospheric detection models, but the long-term solution will be based on the implementation together with other innovative solutions developed in innovative projects. This is the reason of proposing **synergies with other projects** within the CL3 Horizon program, see section 2.

The **main innovation of the PENTION project is the implementation of Vapour Detection technologies for synthetic drug detection**. The next paragraphs will summarise the current state-of-the-art technologies related to vapour detection where it is demonstrated the technical superiority of MION’s technology in terms of sensitivity and selectivity which will be the key to validate the PENTION concept during the project.

➤ State of the Art Vapour detection technologies

It is important to remark that to date, **no Vapour-based technologies have been implemented for synthetic drug detection almost in any scenario**. It is true that some complex analytical techniques are in place for the analytical analysis of seized drugs, but not with the objective of detection, but for the objective of analysis. This is because the current vapour-based technologies of the few competitors do not have the detection performances, in terms of sensibility and selectivity, required for the accurate detection of threats versus the VPSCAN detection instrument³², developed by MION, that has already achieved remarkable milestones at international level. It has pushed the limits of detection to an impressive 0.01 ppq (parts per quadrillion) for identifying explosives within containers [Ref 25], laying the foundations for the development of the proposed in PENTION.

A detailed analysis of the potential competitors is now presented. Companies like **KARSA** (<http://karsa.fi/>) in Finland have developed vapour-based technologies, mainly for explosive screening, but also for other fields like atmospheric analysis; however, their technologies are not ready for implementation yet. According to the official website of KARSA (<https://www.karsa.fi/security/scenthound-evd>), they claim the “product” is able to detect picograms. While this is an impressive level of sensitivity, it may not be sufficient for detecting substances with medium to low vapour pressure as it has been presented before. Other research initiatives aimed at combating the trafficking and smuggling of various threats into airports or seaports worldwide are primarily at the prototype stage. These prototypes are predominantly tested only in experimental settings. While major companies in the detection equipment sector are diligently working on enhancing the selectivity and sensitivity of their instruments, there remain significant gaps in the sampling technologies employed by these systems, mainly the capacity to sample at a high flow rate which is key to minimise the interruption of the logistic flows.

Even though the vapour pressure shown in Figure 3 are not too low, these figures are the vapour pressures of the

³² <https://vpscan-evd.com/>

substances in “open” conditions (saturation level at 25°C), but MION’s extensive experimentation, after the analysis of more than 1,500 sea containers at the port of Vigo (Spain) using VPSCAN technology, has demonstrated that the real concentrations found are notably lower, between 4-5 times lower than the saturation level. This reduction in concentration is influenced by numerous variables, but the main reason is the complex concealment and packaging used by the dealers (see Figure 7) which impede the free vapour emission of the substances. In fact, the estimated vapour pressure **demonstrates the need for detector sensitivities in the order of sub-ppq to be able to detect drugs in complex and ingenious concealments, parcels or vehicles.**



Figure 7 – Examples of drug seized in different concealments (parcels, cars and luggage) (source: Internet pictures)

The key to the outstanding detection performances of MION’s technology is the **work performed during the past 15 years in vapour detection development**. MION’s team has collaborated for over 12 years with relevant organisations worldwide, including TNO and Dutch Customs (NL), Service Technique de l’Aviation Civile (STAC) in France, Defence Science and Technology Laboratory (DSTL) in the UK, Spanish Customs and Guardia Civil in Spain, Bundespolizei, Israeli Prime Minister Office (IPMO) and Transport Security Laboratory (TSL) in the US. The results achieved thus far with VPSCAN, coupled with the extensive expertise of PENTION partners, **hold significant value in shaping and developing innovative approaches to vapour screening**. Collectively, this wealth of experience serves as a foundation in the development of PENTION which represents a pioneering concept in exceptionally precise vapour screening solution applied to one of the main challenges that the EU is facing nowadays.

From 2022, MION has **reinforced the collaborations with some EU customs**, mainly Dutch, Belgium, Spanish and Irish customs authorities for the detection of cocaine in maritime containers ([METEOR EU-funded project](#)). And now this collaboration is in expansion with Portuguese customs and other organizations in Latin America where the problem of cocaine is in the next level. The METEOR running project has allowed the screening of real “hot” containers in different locations for the understanding of cocaine vapour headspace providing a unique opportunity for MION to have access to real cargo concealments of cocaine. Up to the proposal submission, **more than 50 containers have been screened in different parts of Spain and Belgium with quite relevant results in terms of detection**. The current detection routines are based on the detection of the own molecule of cocaine and two degradation products. MION EVD is the only technology in the world that has reported the capacity to detect the molecule of cocaine in gas phase because of the sensitivity. Hydrochloric cocaine, which is the most common way of transportation, has a vapour pressure of around 50 ppt (parts per trillion) which could be detectable by other technologies, however, the concentrations found in real containers are around 50 ppqs which is aligned with the dilution factor we have identified in our extensive container analysis many years ago, and **the capacity of detecting at the level of tens of ppqs is only possible by MION EVD**. More details of the technology are described later on. One example of the detection can be shown in the following infographic.



Figure 8 – Positive detection of cocaine in a container full of zeolites (chemical absorbent) using MION EVD – May 2024

➤ Current status of development and TRL

The PENTION project has the **primary objective of delivering a TRL7 operational PENTION prototype** (Sampling and Analysis innovative technology), that will be tested and validated in real operational environments,

aligning with the proposed use cases. The project is founded on the integration of technologies, each at different TRL, which are explained as follows:

1. **Sampling Technology:** MION air sampling technology is ground breaking because it is one of the few technologies in the world with the capacity to efficiently collect vapours, **no particles**, at high flow rates (up to 350 Liters per minute) which allows for the screening of an entire truck within 3 minutes. This technology has been validated up to TRL7 for explosives and cocaine sampled within contained volumes such as containers or trucks. PENTION will require the validation of this concept for parcel screening for synthetic drugs which has only been tested in simulated environment up to TRL5 (Figure 9 and Figure 10). It is planned to do real experiments, quantifying the vapour emissions of these substances, initially using simulants within different concealments for the characterization of the vapour emissions through different layers of concealment. The challenge here is the development of two sampling units:
 - a. **Fixed sampling unit for parcel screening (PENTION-S)** which is a solution for the static version of the PENTION system. To do this, a variety of experiments have been conducted in the lab, as shown in Figure 9. In this example, analysis of luggage for explosive detection was conducted.

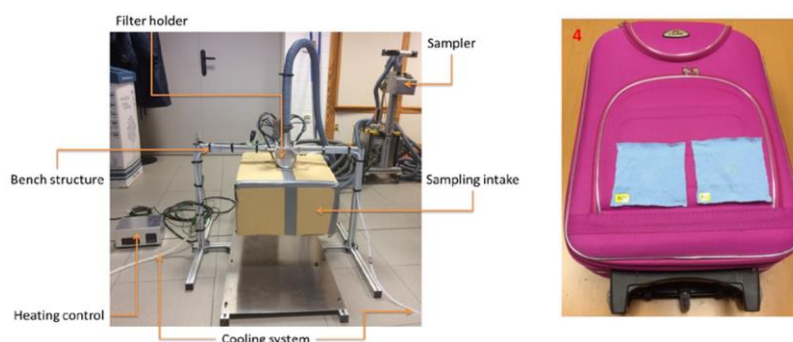


Figure 9 - TRL5 prototype screening chamber for individual bags testing in MION's lab

The results were quite promising and the concept was validated in a more realistic scenario (MION's facilities checkpoint - TRL5) during summer 2023 (see Figure 10) within the VASCREEN project³³ where pieces of luggage were analysed to assess background levels and then small quantities (mg) of Nitro-glycerine (NG) and TNT were concealed in the interior of the baggage (Figure 10-centre). The results of the final testing showed a detection rate higher than 90% for TNT and around 75% for the case of NG for the case of threats and the level of milligrams.



Figure 10 - VASCREEN proof of concept in a realistic environment (MION's facilities control access)

These tests were the origin of the PENTION proposal, specifically for use case #2 (see use cases section), which will comprise of parcel screening applications in the interdiction of drug trafficking.

- b. **Autonomous and wireless portable sampling unit (PENTION-M)**, which is a fully autonomous and portable sampling system that provides more versatility, and is applicable for use case #1 (detection of clandestine laboratories) and #3 (vehicle screening). The current version of vapour collection technology (filter collection + filter-holder technology) is at TRL7, backed by over a decade of rigorous real-world testing in collaboration with the most advanced security organisations, including TNO, DSTL, ICT-Fraunhofer, IPMO, TSL and others, as well as prominent customs authorities such as those in the Netherlands, Belgium, Ireland, and Spain. This technology has been extensively validated for cargo screening applications in fixed screening environments. The development of a wireless/autonomous

³³ <https://securit-project.eu/funded-projects/vascreen/>

version is a challenge for many reasons, PENTION-M will require innovative solutions in terms of air movement, preconcentration and trapping. STAM will be responsible for this development.

- 2. Analysis Technology:** The technology proposed for PENTION is based on MION's VPSCAN system, **primarily at TRL7** for cargo screening (see methodology section for further details) in close environments. However, several improvements and **innovative developments are needed to adapt the technology for the analysis and detection of synthetic drugs** for the three cases defined by Netherlands Police and Limburg Region. A deeper description of the technology will be detailed in the methodology section.

The **final TRL of the complete PENTION technology will be 7** for synthetic drug detection in the different use-cases. Validation will take place in operational environments in different locations with the Limburg region and with the support of NP. The **proposed innovations** of the project are deeply described in the next section. #SPRJ-OBJ-PO\$#

1.2 Methodology #@CON-MET-CM@# ##COM-PLC-CP@#

The PENTION project relies on the development, integration, testing and validation of a new concept of vapour screening solution with a focus on supporting Customs and other authorities in improving their detection capabilities for synthetic drug detection, and the objective to implement it to NPS in the short-term. Initially focused on the three proposed use cases: i) Clandestine Laboratory detection, ii) Synthetic drugs/precursors detection in parcels and iii) Detection of synthetic drugs in vehicles and the three target synthetic drug compounds: amphetamine, MDMA and methamphetamine; however, the detection can rely on some drug precursors and degradation products, in total, at least 6 substances (KPI#3) will be integrated in the detection routines at the end of the project.

PENTION technology relies on MION's VPSCAN which is a technology developed for cargo screening as illustrated in Figure 11. This screening approach is divided into two parts: i) **the air sampling system**, where an air sample is extracted from the target scenario, typically a container, and collected on a trapping filter, and ii) the **Analysis system**, where the collection filter is analysed and the verdict Alarm/No Alarm is given.

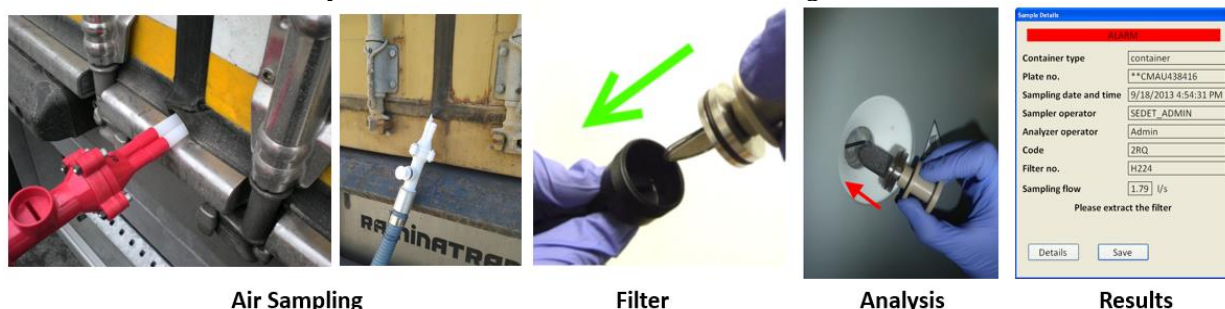


Figure 11 – VPSCAN Vapour detection analysis concept

The two step screening process is required due to **the presence of very low volatile substances that requires the preconcentration of the sample** in a separate sample collection system. PENTION project relies on the adaptation of VPSCAN technology for the screening of synthetic drugs and their detection, which requires **introducing innovative technologies** (see Figure 12).

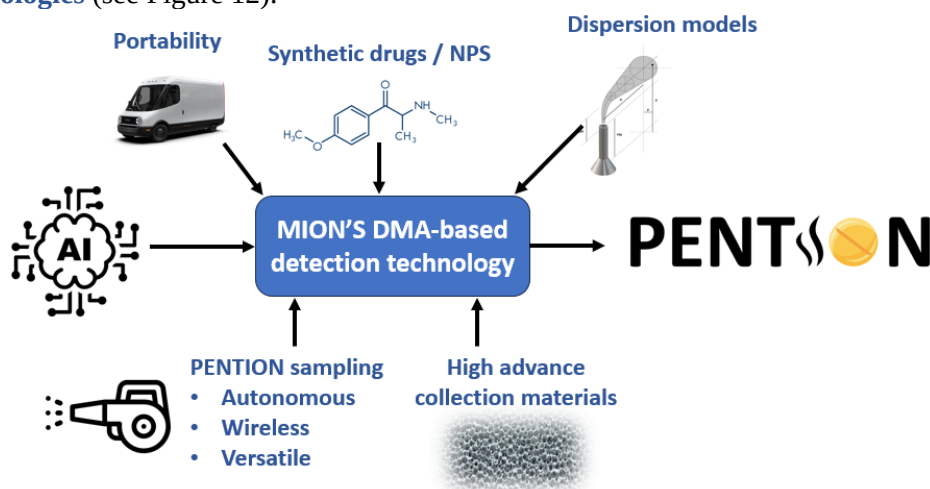


Figure 12 – PENTION main innovations to be implemented

The methodology and the proposed innovations in sampling and analysis are described below depending if they are associated to the sampling or the analysis parts.

1.2.1. PENTION sampling units

Two PENTION sampling systems which are based on MION's vapour sampling technology are being developed. The main principle of the detection is directing airflow towards a **surface covered with a coating** which has a chemical affinity for the vapour(s) of interest. Vapours are retained in the coating which can be analysed at a later date. Figure 13 represents the MION's sampling concept.

The current system for cargo sampling consists of a disposable Teflon tube (1) that is introduced in the sampling target (container, pallet, baggage, etc) and a filter-holder (3) where the trapping filter (2) is contained. The system includes a valve (4) to prevent air from the pump (5), and the exhaust (6) line. The collection filter is based on a metal mesh coated with a chemical adsorbent, which is placed in a filter holder (Figure 14 on the right). The functional coating is Tenax GR, which is a porous polymer resin with 23% graphite to improve the trapping performance. It is a multi-purpose coating with an affinity for alcohols, polyethylene glycols, diols, phenols, monoamines and diamines, ethanolamines, amides, aldehydes, ketones and chlorinated aromatics that traps for most of the volatile compounds (low selectivity). A pre-filter prevents particles from entering the system, as indicated in Figure 14 with the blue arrows showing the direction of the airflow.

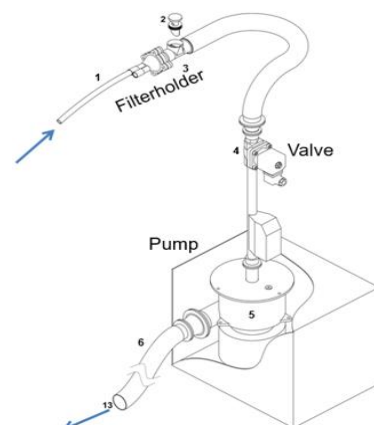


Figure 13 – MION sampling concept

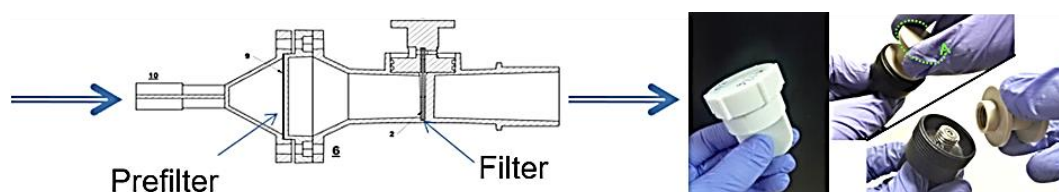


Figure 14 – A state-of-the-art air sampling system (left) and filter-holder (right)

The sampling unit is comprised of a filter, with a new unused Teflon tube attached to it. The air aspiration pump allows a high-volume sample to be taken. Once complete, the filter is removed from the filter holder and put into a special storage container (Figure 14 – centre) to be stored until the analysis. This sample holder has been specially designed for this filter minimizing the amount of air inside to avoid undesired contaminants. Once taken, the filter is then brought to the analyser, where it can be desorbed, and the substances analysed.

The current sampling system works at a flow rate of around 300-350 Liters per minute, depending on the type of scenario. Smaller times for parcel screening and longer times for open area samplings are expected.

➤ Ambition and innovation in the PENTION project for the sampling unit

For the **static version (PENTION-S)**, the following characteristics are required:

- Easy loading/unloading of the items/parcels, with the possibility to install it inside the courier logistics belts,
- Easy and automatic cleaning process, including anti-contamination procedures,
- High throughput,
- Optimal sampling flow distribution to maximise the collecting efficiency,
- Minimal use of consumables to reduce costs, operations and environmental impact.

Various studies have been conducted by MION and STAM to **optimize the vapour collection for the case of** vapour explosive detection. Challenges in detection such as limited soaking time, sampling in a non-sealed volume sampling, non-optimal vapour collection selectivity, limited target signal, and high risk of cross-contamination, are proposed to be solved by the development and integration of the following innovations starting from the current TRL5 version validated at MION facilities. In order to maximise and enhance vapour emissions (sensitivity), the design of the aspiration system will include the following an **optimal design** to optimise the airflow to cross the majority of the surfaces among the different items. Figure 15 illustrates how a wrong sampling flow can collect air from the outside of the chamber and generate false alarms or dilute the sample.

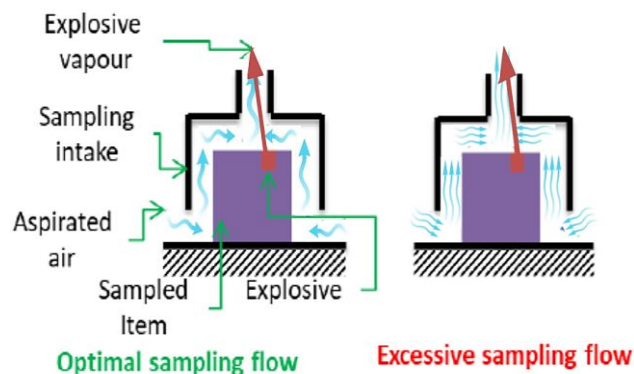


Figure 15 – MION's studies for the optimal vapour collection

One important innovation is the **Capacity for Contamination Prevention and Auto-Cleaning**: Ensuring that the sampling system can prevent contamination and perform auto-cleaning after detecting a positive signal is crucial.

This involves developing mechanisms to maintain the screening area clean by the implementation of special materials or coatings and also, the use of films to protect the critical surfaces where the contamination can be deposited. If an analysis triggers an alarm, the protocol involves activating cleaning procedures removing the disposable parts. Additionally, for the courier screening use cases, the variety of parcels is huge, so it is required to develop a solution with **high versatility**. The adapted technology needs to be versatile and adaptable to logistics flows and scenarios. Different airports and security setups may have unique requirements, and the technology must be flexible enough to accommodate these differences in locations.

For the **mobile version (PENTION-M)**, the following characteristics are required:

- Full autonomy,
- Easy transportation, operation and deployment
- For the implementation of the prediction model, the PENTION-M system hardware will be integrated with a GNSS receiver capable of receiving the positioning signal from the Galileo constellation.
- Suction pump suitably selected for technical performances (flow rate, pressure) and cost-effectiveness (procurement, operation and maintenance),
- Minimal use of consumables to reduce costs and environmental impact.



Figure 16 – Current MION's portable sampler.
Part of the VPSCAN system

The main innovation behind the autonomous sampler is the **capacity to operate without external power supplies and the capacity to be easily deployed in the field** which is a technological challenge. Current MION's sampler has an electrical demand up to 1.2 KW and it has a total weight around 12 kilograms (see Figure 16). Additionally, the sampler provides vibrations and a lot of heat, close to 65°C. The challenge for PENTION is to develop a new sampling unit with the same performances in terms of sampling flow and pressure, but is a portable, autonomous and easy to transport instrument. Technology behind batteries has improved a lot during the last years and this will be the key for the proposed development. Something common to the two sampling versions is the improvement of **vapour collection selectivity**. The current sampling technology, including absorbent materials, prefilters, etc are optimized for the collection of vapour explosives, and currently MION is working on its optimization for cocaine detection within the METEOR EU-funding project, however, the chemical structure of the synthetic drugs is quite different and it requires the use of new more selective absorbent materials which is an important innovation in the project, some materials have been already selected such as polymeric-based solutions or active materials such as MOFs (metal-organic frameworks) or activated carbon, but a deeper analysis of the literature will be done during the initial stages of the project.

STAM, as technical developer of the PENTION sampling unit, will start from the design requirements outlined by MION (technical), NP and PENTION AB (operative) and identify the main engineering decisions, related to several conceptual designs. The advantages and disadvantages of each solution will be evaluated and scored against criteria that have to be identified as well. This will allow designers to identify the most suitable solution in a trade-off analysis. Afterwards, the conceptual design will be elaborated, and a prototype will be constructed, with the goal of testing the main functionalities and confirming predictions and design choices. Finally, the optimised prototype will be thoroughly tested with NP and Limburg region end-users to assess compliance with requirements, specifications and real needs.

For the detection of clandestine labs, it is needed the implementation of **atmospheric dispersion models** based on AI. With a predilection towards open-source intelligence and open sensing, air samples taken will be sampled and depending on sampling results and connected indicators (e.g., concentration of the target substances in the air) and other controllable factors (e.g., weather conditions, surrounding structures), dispersion gas models **will estimate the position of the source of the emissions**. Gas models will be engineered featuring at least four components³⁴: (1) **Data Assimilation and Feature Extraction**: ML shall process sufficiently large volumes of environmental and meteorological data, such as wind speed, temperature, humidity, and topographical details, to identify patterns affecting dispersion – this shall harness state of the art approaches such as OpenFOAM or ANSYS Fluent for CFD-based data generation as well as hybrid models combining traditional atmospheric equations (e.g., Gaussian plume

³⁴ Pierre Lauret, Frédéric Heymes, Laurent Aprin, Anne Johannet, Atmospheric dispersion modeling using Artificial Neural Network based cellular automata, Environmental Modelling & Software, Volume 85, 2016 Pages 56-69, ISSN 1364-8152, <https://doi.org/10.1016/j.envsoft.2016.08.001>

model) with ML³⁵ (e.g., LSTMs or GNNs) to predict dispersion patterns; (2) **Parameter Estimation**: Traditional dispersion models rely on empirical parameters, which might be difficult to determine precisely. ML algorithms, such as regression or neural networks, shall be used to optimize these parameters by training on historical dispersion data, e.g., by using GIS software to map and overlay detected anomalies and predicted source locations; (3) **Real-Time MLOps Adjustments**: ML models shall be rigged to continuously update dispersion predictions based on incoming sensor data, allowing for real-time adjustments; (4) **Uncertainty Quantification**: specific fine-tuning ML techniques (e.g., chemical Bayesian networks) shall be used to estimate the uncertainty of predictions, providing confidence intervals around dispersion outcomes.

1.2.2. PENTION Analysis Unit.

The PENTION project is based upon an innovative analytical instrument for the collected samples analysis. This instrument is a combination of MION's DMA (Differential Mobility Analyser) technology and a mass spectrometer as the primary detection method. Here are the key components and features of this analytical instrument:

- **DMA Technology**: One of the key technologies included in the PENTION is MION's DMA (see Figure 17.a) which is a specific technology related to Ion Mobility Spectrometry (IMS), that classifies molecular ions based on their electrical mobility in a gas. The DMA has a planar geometry, which has demonstrated resolutions of 110 (1/FWHM) and close to 100% ion transmission (compared with 1% of other competing technologies based on time-of-flight separation^{36, 37}). A non-trivial aspect that makes MION's DMA unique is that the sheath flow is **kept highly laminar** at very high Reynolds numbers (>100.000), which gives the DMA its unmatched resolution and transmission performance. When an ion goes into the DMA, it is carried horizontally (see Figure 17.b) due to the gas, however, an electrical field orthogonally forces the ions to generate different trajectories according to their electrical mobility. This technology is already at TRL7 and no modifications are planned during the project.

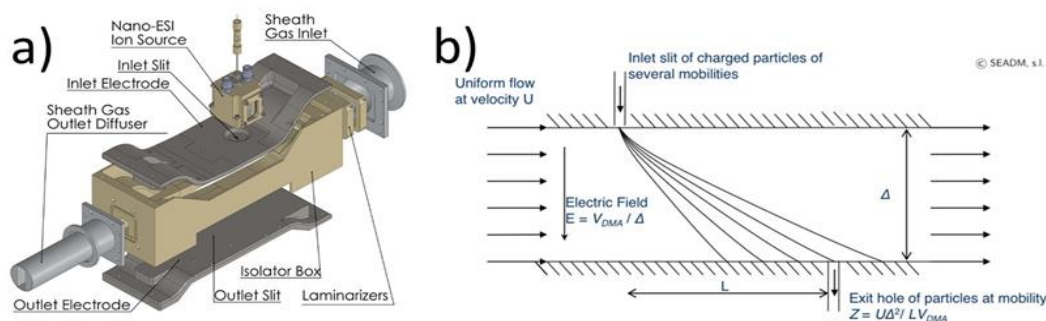


Figure 17 - a) DMA, as MION's core technology. b) Principle of operation. Note the two orthogonal fields (electrical and flow)

- **Mass Spectrometer (MS)**: The MS serves as the main detection method. Mass spectrometry is a powerful analytical technique that can identify and quantify various compounds within a sample based on their mass-to-charge ratio. Specifically, a TripleQuad unit from Sciex (<https://sciex.com/>) is planned to be used for the project, MION will provide this instrument with the only cost of the maintenance contract for the whole duration of the project. This technology is highly versatile and widely used for substance analysis. A challenge for the PENTION project is the embarking on the VPSCAN analyser in the vehicle.
- **Thermal Desorber**: A new thermal desorber will be developed as part of the PENTION project. This desorber is designed to work in conjunction with a **new vapour-collecting filter** aiming to optimise the release of vapours from the collection filter. This optimisation is achieved by using a desorption temperature ramp, which minimises the total analysis time, which is this one key objective.
- **Ionisation Process**: Before the collected vapours can be analysed by the DMA and MS, they need to be ionised. MION has already developed its own Atmospheric Pressure PhotoIonization (APPI) sources which will be used in the PENTION project. MION has also registered the trade mark (SEFIMAX) and it provides the required ionization capabilities for the PENTION project.

³⁵ Theodosios Kassandra, Evangelos Bagkis, Lasse Johansson, Yiannis Kontos, Konstantinos L. Katsifarakis, Ari Karppinen, Kostas Karatzas, Machine learning-assisted dispersion modelling based on genetic algorithm-driven ensembles: An application for road dust in Helsinki, Atmospheric Environment, Volume 307, 2023, 119818, ISSN 1352-2310, <https://doi.org/10.1016/j.atmosenv.2023.119818>.

³⁶ Amo-Gonzalez, M., Fernandez de la Mora (<http://pubs.acs.org/action/doSearch?ContribStored=Fernandez+de+la+Mora%2C+J>). Juan High Dynamic Range Differential Mobility Analyser (DMA) Coupled with a Mass Spectrometer and several Ionization Sources (Pending publication at the Journal of the American Society for Mass Spectrometry, 2016)

³⁷ Amo-González M., Pérez S. "Planar Differential Mobility Analyser with a Resolving Power of 110".

The combination of DMA technology, mass spectrometry, the thermal desorber, and optimised ionisation techniques makes this analytical instrument a versatile tool for detecting and analysing a wide range of substances at very low concentrations (sub–ppq level) to meet the specific needs and objectives of the PENTION project in terms of substance detection, analysis speed, and sensitivity. **This combination will be the differentiating factor of PENTION analysis technology from potential future competitors**, if any.

➤ **Ambition and innovation in the PENTION project for the analysis unit**

The aim of the PENTION project is to adapt MION's technology to the case of synthetic drug analysis which is a technological challenge that we have broken-down into these 3 technical dares, depending on the use case:

- The main activity is the development of the **detection routines for the proposed three synthetic drugs, and precursors/sub-products, in the gas phase**. One of the main characteristics of synthetic drugs is that they are not available in the nature and a chemical processing is required starting from **some precursors** which is very positive from the detection point of view since the precursor detection allows the detection in different parts of the manufacturing chain. Figure 18 represents how the detection of precursors can allow the detection from the initial supply to the manufacturing, even it is possible in the distribution because some traces can be attached in the final product. The detection routines proposed for PENTION will include both final drugs and precursors.

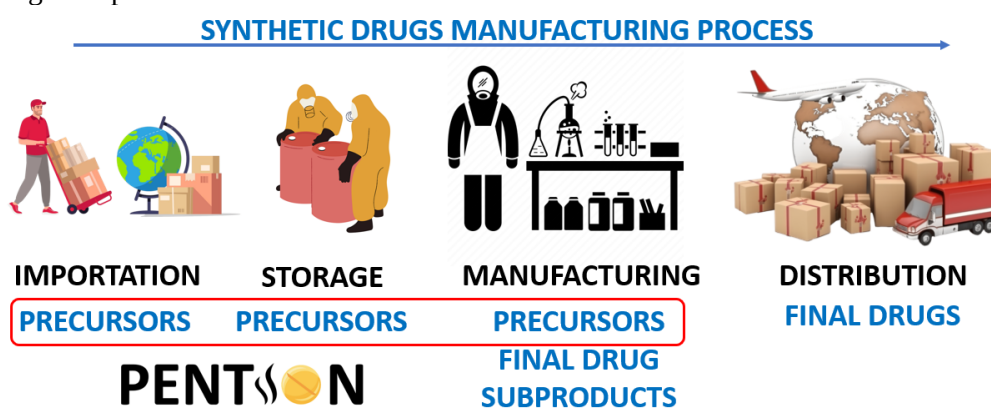


Figure 18 – Synthetic drugs manufacturing process, from supplier to final distribution

Even though the detection supported by precursors is positive, the variety of precursors is wider than the drugs and the type of precursors vary. For example, for the manufacturing of amphetamine and methamphetamine, there is a big list of precursors such as AIBN (methyl thioglycolate), Ammonium formate, benzaldehyde, benzylcyanide, BMK, DEPAPD, ephedrine, formamide, formic acid, glycidic derivatives, iodine, MAPA, nitroethane, tartaric acid, etc.

- **Portability** of the PENTION analyser. For the use-case #1, it is essential to have results in situ and apply the dispersion models on real time. Additionally, the operation of drug manufacturers is quite fast, and it is possible to dismantle a clandestine lab in less than one day, so it is mandatory to have the complete PENTION system mounted in a vehicle which is a technological challenge. STAM will work on this and it is proposed to be mounted in a van with the capacity to screen large areas in some hours (KPI# 13).
- **Analysis time is critical to minimise the impact on the parcel logistics**. The current analysis time for cargo screening is in the order of 5-7 minutes. This includes the preparation of the thermal desorber conditions to the final data processing plus the cleaning process after the analysis. This time is not aligned with the parcel process, so we aim to reduce the **sampling time to less than 1 minute** for this use case and around **2-3 minutes for the other use cases**. To do this, an optimization of the desorption process will be done using the new desorber adapted to the new collection filter. In case the proposed analysis time is not reachable (Risk#4), the mitigation action is the replacement of the multi-temperature desorber with a GC (Gas Chromatography) unit.

- Intelligent identification of interferent and peaks.** The variety of items in parcels and in open areas is much different from the case of air/maritime cargo, so the presence of potential interferents much higher. In the following picture (see Figure 19), we can see the case of an interferent in the TNT (Trinitrotoluene) from a real detection in a container. The raw data includes the information from an interferent and the real TNT, after the application of MION's peak deconvolution process, it is possible to separate the signals from the interferent and the real TNT which has an intensity of around 30 fg (femtograms). The current Gaussian peak deconvolution process has been developed using a semi-manual "trial and error" process and it is only valid for some specific detection channels in negative ionisation mode, furthermore, it is not possible to be automatically extrapolated to new detection substances as proposed for PENTION because each detection channel has specific interferents.

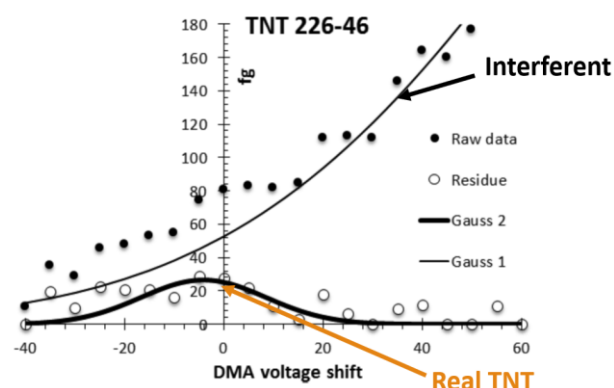


Figure 19 - Real detection of TNT and interferent using current semi-manual peak identification techniques. X-axis represents the mobility in Volts, 0 Volts is the target mobility. Y-axis represents in the intensity in femtograms

- In addition to this, most of the substances to be detected in the PENTION project (synthetic drugs) are ionised in positive mode, versus the traditional explosives based on nitrate compounds which are prone to be ionized in negative mode. The detection routines in positive mode are not highly developed yet and they have associated an important complexity in terms of chemical background noise. Thus, it is imperative to automate the process of peak and interferent detection. To solve this problem, it is essential to implement an **AI-based, self-optimising processing technique harnessing the combined power of Deep-Learning (DL) with AI-driven Machine-Learning Operations (MLOps)**, with the capacity to learn from previous analyses and, given pre-specified or learnable conditions, trigger a self-optimising loop. In addressing this gap, USN shall devise SIT (Self-optimising Substance Intelligence Tool). SIT shall use artificial intelligence approaches to compound the semi-manual analysis currently used to address substance identification and general intelligence. USN will develop such specific AI-based tooling and pipelines to enhance the peak identification capabilities of the system by the intelligent fitting of the different parameters in the inputs, both peaks with mobilities in the target mobilities or interferents (e.g., substances in the gas phase with similar properties as the target molecule that can interfere in the measurement) across different known degrees of freedom. The implementation of SIT through DL/MLOps techniques during the project will follow an Action Design research approach³⁸. Specifically, before the development of said techniques, USN shall enact a systematic selection of the most suitable AI-based and MLOps methodologies. Then USN shall evaluate within the scope of a continuous research experimentation loop, a comparison between the incrementally-defined AI solutions in PENTION and the pre-trained semi-manual model(s) traditionally utilised by MION as a baseline. Finally, SIT shall integrate components for self-optimisation based either on heuristic or predictive approaches to continuously fine-tune its operational behavior against real-life working conditions. In addition to the more accurate identification process, SIT shall provide additional proven software quality characteristics: i) The algorithm will be robust regardless of the level of internal background noise and potential interferents in the input data (Reducing FAR and increasing DR), ii) Reducing the sampling times (KPI#1a and b), iii) Integration of the model into MION's interface, iv) Generation of visual reports for additional analyses and understanding, and v) Friendly interface to be operated by a non-trained operator.
- Additionally, and thinking about mid-term, the **study of the vapour pressure of NPS** is proposed. As described previously, NPS are a real and growing problem globally, but the detection of these substances has some additional challenges. The main one is the low vapour pressure (volatility) associated with the low amounts transported, which makes detection much harder, or even impossible for imaging detection instruments. In addition to this, no solid technical studies are available currently as explained before, this is why the RFI (Request For Information number 70RSAT24RFI000001) released by the US government in November 2023 calls the international community to provide "*R&D efforts to disrupt the fentanyl supply chain*". **The challenge here is the determination of fentanyl vapour pressure** using MION's technology and defining the requirements to be detected by vapour technologies, perhaps being the only technology with the capacity to experimentally determine the vapour pressure and consequently, detect it. Within the scope of the PENTION project, it is planned to establish cooperation with the US government (see section 2) in terms of exploitation and commercialization of PENTION technology.

³⁸ <https://misq.umn.edu/action-design-research.html>

➤ PENTION use cases

The specific use cases and scenarios for validating the technology will be meticulously outlined in T1.2 during the project's initial months. While a conceptual framework for demonstrative use cases has been presented by experts, and end users, mainly after the initial conversations during the proposal preparation with **Netherlands Police and Limburg region**, which are overwhelmed by the number of clandestine labs in the region with the capacity to produce amphetamines for the rest of the world, and the **Spanish National Police and Spanish Customs (associated partners)** and **Guardia Civil (member of the AB)**, which are also concerned by the increasing number of seizures of new drugs in different parts of Spain, specifically in Ibiza (Balearic Islands). The result is the formulation of **three preliminary use cases**. These cover a number of real-world concealment and detection scenarios, where the developed technology can be deployed for the detection of illicit materials and their precursor chemicals. The use cases will be demonstrated within different infrastructures and scenarios as it is defined in the table below and they are scheduled for the second half of the project, initially at MION facilities in WP3 (PENTION mobile-lab Integration and in-house testing) and also in WP5 (validation, testing and pilots) in the end-user region and facilities.

To validate the technology and significance of the data/results generated during testing with real substances, an extensive background study involving realistic screening and detection scenarios will be conducted (without target substances). These tests aim to understand the prevalence of complex background matrices and their impact on detection performance. Accuracy measurements and optimisation will be made to fine tune the detection algorithms remove the incidences of false alarms and incorrect detection being made.

The proposed real-world concealment and detection scenarios (the use cases), will serve the following key objectives:

- Validation of the **PENTION sampling system** in operational environments (TRL7), clandestine labs detection, vehicle screening and parcel/mail screening,
- Validation of the **PENTION detection capacity**,
- Preliminary determination of the **FAR and DR of PENTION** technology,
- Determination of the background signals of the different target substances in different configurations for the **training and validation of the detection models**.

The proposed use cases are a reflection of current and pressing challenges in the detection of these substances. It is essential to recognize that the selected use cases are not constraints of the technology but have been chosen by real experts and end users in the regions where the illicit trafficking occurs. In addition to the end users, technology experts mainly ITI, which has been working on the NPS detection since last decade in Canada and the US, will provide their feedback for the proper implementation of the use cases. **Canada is currently suffering a huge problem with synthetic drugs**, even harder than in The Netherlands. During the proposal preparation (01/11/2024), the Canadian police dismantled what they called the **most sophisticated illicit drug "super lab" in the country**, saying they had seized "a record number of illegal firearms, synthetic drugs and precursor chemicals."³⁹, with a big seizure of MDMA and methamphetamine. Canadian TV alerts the population about the production of synthetic drugs and their risks (<https://www.cbc.ca/player/play/video/9.6560068>). ITI has been working for more than 30 years on the development of technology for the detection and identification of synthetic drugs and its support will be essential for the project implementation and bringing high expertise to the EU.

The use cases exemplify the proposed solution's effectiveness in addressing the challenges faced by Border and Customs authorities, police authorities and security officers who encounter such illicit material concealment scenarios. The three proposed use cases aim at covering the whole supply chain: Early Detection (production), Transportation hubs and Final distribution (vehicles). They are described as following:

Use case 1: A clandestine laboratory of new drugs is hidden in a rural/industrial area
Objectives: • Validation of PENTION technology in a realistic scenario • Quantification of the vapour emissions typically generated in the clandestine laboratories • Study the vapour dilution factor and impact of sampling distance, time and weather conditions. • Study of potential interferences and the background signatures in the air • Ability of subtracting upwind vapour signature from suspected area signature
Infrastructure: 1) MION facilities, 2) Netherlands Police resources, 3) Limburg region in specific areas, and 4) standards of substances and precursors.
Hardware: PENTION-M prototype

³⁹ <https://www.cbsnews.com/news/canada-drug-super-lab-busted-fentanyl-guns/>

Challenge: The validation of the proposed PENTION system, when operating in such an open area for sampling, requires extremely high levels of sensitivity and selectivity, due to the number of potential interferents present at very low concentrations. Even though the detection potential of PENTION technology is enough for the successful identification of the target illicit materials, the detection of Volatile Organic Compounds (VOCs) and semi-VOCs during the manufacturing process within the clandestine laboratory environment is relevant, as has been referenced in previous sections. Additionally, the drug production generates a lot of “synthetic” waste which is a big environmental problem with a costly cleaning procedure, but it can be used for detection and it will be analysed during the project.

The objective is the detection of vapours from amphetamines production at different levels and estimate the detection capabilities of the PENTION technology. Initial data collection of backgrounds will take place at MION facilities, with the support of Spanish Customs, Guardia Civil and Spanish Police, and positive samples (with threat) in open environments at MION facilities and also in the Limburg region, supported by Netherlands Police.

Solution: Validation of the technology in an open-air environment under different scenarios, simulating the manufacture of illicit drugs whilst detecting the precursors chemicals and volatile biproducts of drug manufacture.

Use case 2: A box with precursors of synthetic drugs (Amphetamine) is coming from a foreign country and it is detected in the courier warehouse / mail offices / airport checkpoint

Objectives:

- Validation of PENTION technology for detection of precursor chemicals, some of which maybe “legal”
- Validation of PENTION technology for identification of shipment (package/mail) that contains a narcotics concealment and its application in courier/mail scenario
- Setting a detection threshold for a list of substances in moderate quantities under realistic scenarios
- Vapour pressure study of the different substances/precursors

Infrastructure: 1) MION facilities simulating different packages, 2) Amphetamine precursor samples, and 3) Limburg region in warehouses

Hardware: **PENTION-S** prototype

Challenge: Limburg region is the main new drug producer in the EU. The procedure is based on the importation of the precursors, the manufacturing of the drugs and the exportation to the final destination, e.g., to Ibiza⁴⁰ (Spain). The ongoing challenge is the detection of precursors before they reach the manufacturing lab, through the screening of suspicious parcels. PENTION technology has the capability to detect chemical volatiles without the need for opening it. Some precursors of amphetamines have been widely studied, such as the BMK (benzyl methyl ketone), but other potential precursors will be selected at the project beginning. The challenge is the detection of some of these precursors in a realistic environment simulating real packages and real concealments, with a focus on the minimalization of false positives.

Solution: Once the detection procedures and routines are developed and validated, it is planned to do extensive background testing to know the typical background levels in different scenarios and under different conditions. After this, detection tests will be performed in different configurations of parcels. This solution has a direct application to many scenarios such as courier distribution centre (e.g., Venlo Distribution area) or airport checkpoints (Beek airport in the Limburg region).

Use case 3: A box with MDMA is detected in a vehicle at a border crossing point (checkpoint)

Objectives:

- Chemical characterization of MDMA vapour headspace
- Experimental validation of the vapour pressure of MDMA
- Identification of the MDMA sample once contained within the box inside different positions in a vehicle
- Adaptation of the PENTION solution to vapour vehicle screening

Infrastructure: 1) MION facilities simulating vehicles containing concealments of MDMA, 2) Limburg region in border crossing points and checkpoints, 3) samples and precursors.

Hardware: **PENTION-S** prototype

Challenge: Public transport and vehicles are one of the methods to transport narcotics and their precursor materials as it is described by Zhao, M. in its research⁴¹. In addition, vehicle screening, beyond the use of image technologies, is an unsolved challenge nowadays, this is demonstrated by the [HORIZON-CL3-2024-BM-01-04](#) topic, which is looking for solutions for “detecting threats in vehicles” and the Security UK Innovation funding program [DASA](#).

⁴⁰ C Lora-Tamayo, T Tena, A Rodríguez, D Moreno, J.R Sancho, P Enseñat, F Muela, “The designer drug situation in Ibiza”, Forensic Science International, Volume 140, Issues 2–3, 2004, Pages 195–206, ISSN 0379-0738,

⁴¹ Zhao, M. (2022). The Illicit Supply of New Psychoactive Substances Within and From China: A Descriptive Analysis. International Journal of Offender Therapy and Comparative Criminology, 66(5), 495–516.
<https://doi.org/10.1177/0306624X19866119>

which is also requesting solutions for the same objective. The challenge here is the ability to capture enough material through the vapour sampling of the car, to then be able to identify a concealment of MDMA in a box, detecting the compounds at very small concentrations, including the possibility to use chemical precursors. To do so, a car screening protocol will be defined based on the sampling of 3 areas: i) interior air, ii) fuel tank, and iii) bottom parts. These results can be easily extended to other threats, such as explosives, NPS or chemical threats which are beyond the scope of PENTION project.

Solution: MION technology operates within the range of vapour pressures documented for MDMA, and even lower, and it is planned to calculate the MDMA vapour pressure using commercial chemical standards (samples have been requested for purchase due to slow turnaround times) and the dilution factors with the final implementation in the use case of detecting it in a vehicle.

To implement the different use cases, chemical standards/precursors are planned for purchasing, for example [MDMA at sigma Aldrich](#). MION has long experience working with these substances; therefore, no issues are expected on this matter (see Risk #6 in risks sections). In addition to this, illicit materials and their precursors will be sourced from real seizures for their use in the project which will be provided by the project partners and associated partners.

The PENTION use cases align with the project's scope and adhere to the priorities outlined by law enforcement officers and customs authorities. Nevertheless, it's crucial to note that the outcomes achieved throughout the project possess the potential for other applications and scenarios, demonstrating the solution's remarkable adaptability. The ultimate implementation of these field cases will be executed during the technology validation (WP5) where a comprehensive test plan will be elaborated to conduct the operational tests.

The PENTION project will develop and demonstrate vapour detection technologies for synthetic drug interception, control and inspection, advancing the technology to TRL7. **One of the key outcomes of the PENTION project is the quantitative results of the vapour emissions generated by these drugs in real-world scenarios**; this information will be essential to determine the requirements of future detection systems relying on vapour sampling and detection. For the PENTION project, which is a project with limited resources and budget, the most advanced vapour detection instrument based on DMA-MS will be quantitatively validated. However, the results can demonstrate that it is possible that a less demanding analyser (cheaper/smaller) can be used, in terms of sensitivity; this would positively impact the business plan.

➤ **Interdisciplinarity of the consortium**

The PENTION project benefits from previous cooperation under Horizon between the **SME EU industry (MION and STAM)**, supported by one of the main international players working on Vapour/Trace detection (**ITI**) delivering technologies in America and Canada for more than 30 years. The development proposed at PENTION will be supervised by **PPC**, a company located in the UK, whose founder (Samantha Ollerton) has over two decades of experience in shaping the global security market; renowned for her expertise in threat detection, she has cemented her reputation as a globally recognized authority in the field and she, personally, will oversee the results of the project to provide the best exploitation plan for the rapid market introduction of PENTION technology. The consortium is completed with the **University of Sannio** in Italy, especially the team of the Prof. Dr. Willem-Jan van den Heuvel which will support the industry players for the data analytics in the decision-making processes during the detection procedures. It is quite relevant the active participation of the **Netherlands Police** authorities and the **region of Limburg** (associated partner) in The Netherlands where the problem of clandestine labs and new drugs is one of the main priorities as it has been largely reported in the news⁴². The region of Limburg (RIEC Limburg) is an association that includes, next to the national police, first responders, 35+ municipalities, customs and tax authorities, the active involvement of this region is essential for the implementation of real-world use cases, the active involvement of this region is essential for the implementation of real-world use cases.

Together, they have assembled a **complementary qualified consortium mainly based on SMEs**, covering the following key disciplines: analytical instrumentation for gas/vapour analysis (MION), high-tech mechatronic designs and engineering (STAM), new drugs and detection (ITI), exploitation and business plan (PCC). These skills are required to successfully complete the PENTION project.

Table 6 - PENTION project consortium skill matrix - Interdisciplinarity.

Domains	MION	STAM	ITI	PCC	USN	NP
Partner #	1	2	3	4	5	6
Analytical Instrumentation	X	X	X			
Vapour analysis	X		X			
Engineering	X	X	X		X	

⁴² <https://www.vrt.be/vrtnws/en/2024/06/21/large-drug-laboratory-discovered-in-a-marl-cave-limburg-province/>

New drugs	X		X	X	X	
Detection / customs	X		X	X		X
Testing	X		X	X	X	
Systems Integration	X	X	X			
Border Control			X	X		X
Legal/ ethical issues		X		X		
Security issues	X			X		X
Dissemination	X	X		X	X	
Industrial exploitation	X	X	X	X		
Pilot cases	X		X		X	X

It is quite remarkable at this point, the possibility to include **international partners** in EC Horizon projects since it provides the capacity to **bring international innovation to the EU**. For the PENTION project, the possibility to integrate ITI (Innovative Technologies INC) in the project is outstanding, since we will benefit from the long experience of its founder Sabatino Nacson, who has an experience of more than 30 years in drug detection.

➤ National and international research & innovation activities linked to the project

We have identified key projects with activities closely aligned with the PENTION project scope and have outlined how the outcomes can contribute to and enrich the project's execution (Table 7). In these identified projects, one or more partners are actively engaged, serving as a crucial bridge between these initiatives and our project. These partner-involved initiatives will serve as interfaces, fostering synergies such as the exchange of best practices and cross-fertilization of technical implementation approaches, all within the framework of Key Area 4 (Joint Developments / Collaboration - Conducting Dissemination and Exploitation Activities).

Table 7 - Relevant (ongoing or recently completed) European and international projects.

Project	Partner	Project description and cross-fertilisation opportunities
METEOR	MION	The METEOR project aims to develop a new concept of a portable vapour scanner for container screening based on a multidetector-DMA for customs.
VPSCAN	MION	VPSCAN project aims at the development of a DMA-MS screener for air cargo screening and its certification at ECAC level for aviation security
VECTIN	MION	VECTIN project aims at the development and adaptation of the DMA-MS techniques for container screening
SAUST	MION, STAM	The SAUST project aims to develop a new concept of cocaine detection in aviation, specifically in bags at the arrivals.
TERAEYE	STAM	TERAEYE aimed at developing a sensor to detect explosives based on emissions in the terahertz range, for airport security applications
STREAM	STAM	In the STREAM project, STAM upscaled an IoT wearable device to monitor the safety and security of railway operators working in dangerous environments
BAG-INTEL	STAM	The BAG-INTEL project will provide robust AI-based information utilisation and decision support tools, within the context of advanced detection systems to support customs for increased effectiveness and efficiency of the customs control of air traveller baggage in inland border airports, while minimising the human customs resources needed.
Sampling and testing drugs for technology development	ITI	• Sampling and detection of drugs in Colombia, • Sampling and detection of cocaine in marine containers in Belgium 2022, • Testing and evaluation of vapour sampling for explosives in marine containers, TSA facility in Atlantic City, • Sampling of explosive vapours in cars, trucks and container coming to Parliament Hill, Ottawa, • Sampling and detection of explosives in sport avenues (Rogers Center in Toronto), • Sampling of cocaine bricks hidden in containers in Bahamas, • Drug sampling and detection with CBSA, Canada in Vancouver.

Developing innovative detection products for counter terrorism and counter narcotics	PPC	• Development and commercialisation of Itemiser 5x product for Rapiscan to screen air cargo passengers and baggage for traces of explosives • Development and commercialisation of itemiser 4DN product for Rapiscan to screen air/sea freight, cargo passengers and baggage for traces of drugs • Development and commercialisation of MPCAD mass spectrometer product for Rapiscan to screen for traces of CWA • Development and commercialisation of ETD-100 product for Greyscan to screen air/sea freight, cargo passengers and baggage for traces of drugs and explosives
COSMOS	USN	The COSMOS H2020-funded project offers a variety of predictive and pre-emptive approaches to open, and closed, source intelligence in the scope of cyber-physical systems, that is systems that include sensing as well as actionable hardware like drones. This context is very much consistent with that of PENTION and hence related baselines will be considered for extension/further exploitation here.
NarcoView	NP	The NarcoView ISFP-funded project aims to strengthen the fight against the environmental crime of dumping narcotic waste by gathering improved cross-border intelligence through the development of deep learning algorithms for the analysis of satellite images and use of drone-based remote sensing technology.
Drugsdrone	NP	The Drugsdrone project, funded at Dutch level, aims at the implementation of detection of new drugs with e-nose technologies and how they can be embarked in unmanned vehicles.

➤ Gender Dimension

The PENTION project, while fully aware of the importance of gender equality and diversity, wants to make clear that its central mission is rooted in enhancing new detection methods for new drugs which has a direct impact on customs, but also an indirect impact on security, economical and health. The project consortium acknowledges that gender equality is a fundamental societal goal and that the direct relevance of gender considerations to its core objectives may be limited.

- **Gender Equality Commitment:** The project consortium is committed to the promotion of gender diversity and equality within their organisations and workplaces. The consortium recognises the broader significance of creating inclusive and equitable work environments.
- **Broader Impact:** Although the project's primary objectives may not directly address gender equality, the outcomes and impact of improved customs inspection processes have broader implications. More efficient and effective customs inspections contribute to a safer and more secure environment, which can benefit everyone, regardless of gender.

PENTION consortium's contribution to gender balance and diversity, such as:

- **Diverse Workforce:** The project consortium will actively promote gender balance at all levels of the project, from research and development to management, fostering an inclusive and equitable work environment.
- **Gender-Neutral Dissemination:** Project dissemination information and results will be presented using gender-neutral language and inclusive imagery; to be applied to project reports, promotional materials, and presentations.

➤ Open science practices

Open science will serve as the foundational principle driving communication and dissemination activities within the PENTION project, and it will be seamlessly integrated into every stage of our endeavour. Our approach strictly adheres to the open access policy of the Horizon Europe program, embodying '*as open as possible, as closed as necessary*', due to the nature of the project focus. This commitment is carried out by the provision of immediate online access to scientific information that is both freely accessible and reusable to the end-user.

To ensure the widest possible dissemination and accessibility, pre-print versions of all articles will be deposited in multidisciplinary **open-access repositories** such as Arxiv or Zenodo. Once articles are officially published in scholarly journals or conference proceedings, their final versions will be expeditiously posted to the Institutional open repository. This repository will automatically feed into OpenAIRE, maximising outreach to researchers, policymakers, and businesses. Scientific articles stemming from the PENTION project will be published in peer-reviewed, gold open-access journals accredited by the Directory of Open Access Journals (DOAJ).

Furthermore, **each partner commits to ensuring open access to all deposited publications** via the repository within

six months of publication, assuming an electronic version is available without extra charge via the publisher. Partners will facilitate access to bibliographic metadata, which includes vital details under the grant agreement, the action's name, acronym and grant number, publication date, and any applicable embargo period length, along with a persistent identifier. All scientific peer-reviewed articles will be released under the Creative Commons CC BY 4.0 licence, preserving authors' copyrights. Simultaneously, partners will retain their copyrights and grant appropriate licences to publishers, aligning with Creative Commons licences.

STAM, our dissemination leader (see WP5), will create the dissemination strategy from the project's inception, with a dissemination plan available by M3. This strategy encompasses **early and open sharing** of research findings, including pre-registration, registered reports, and preprints. It also encompasses **research output management**, including research data management as detailed in the Open Science section, and measures to ensure the **reproducibility** of research outputs. While we aim to make as many deliverables public as possible, it's important to note that certain deliverables related to the exploitation of results, security, and business aspects of the developed solution may not be disclosed.

Security in the information. The PENTION project will generate sensitive and classified information due to the own nature of the project, the proper measures to protect the classification of the information will be implemented by the SAB (Security Advisory Board) which will be coordinated by the PSO (Project Security Officer), Mr. Daoiz Zamora from MION. See Security annex for further details

➤ Data management and management of other research outputs

Data management activities, a part of WP1 (T1.4), will start with the formal release of the Data Management Plan (D1.2/D1.3) at M6 with an updated version on M24. Our project will diligently adhere to the open data management guidelines outlined in the Horizon Europe Programme. Specific actions include:

- The project will deposit the research data in a research data repository that facilitates linking publications and underlying data through persistent identifiers and data citations, such as the [EPOS data and service repository](#).
- The project will enable third parties to access, mine, exploit, reproduce and disseminate (free of charge for any user) this research data by attaching Creative Commons Licences to the data deposited.
- The project will provide information via the research repository about the tools available to the users that are needed to validate the results (e.g., specialised software or software code, algorithms and analysis protocols). Wherever possible, the project will also provide these instruments.
- To support knowledge dissemination and impact creation, we will follow a policy that the project deliverables and related datasets will be freely published on the project website that will be created by M24, as long as they are labelled as "Public" through the definition of the governing data licences and as provided in project deliverables table. A publishable version of the document will be produced and shared via the same mechanisms in all other cases.

To support future exploitation of the data either by project partners or by external organisations where open access is implemented, the project will use the following data management principles ([Table 8](#)).

Table 8 - Data management methodology and tools.

Data Summary	Data purpose: The data collected during the project will be used to refine the design of the PENTION solution and ensure that its features are up to standard, fulfilling the needs of the end-users. Whenever it is possible, data will be placed in open repositories. This excludes sensitive/classified data, as well as data about the exploitation of the innovation.
	Data origin: The project data will come from laboratory tests and also from field tests carried out with the assistance of the end users and the PENTION AB. We will sign formal open data-sharing agreements with these customs for the data that can be placed in open repositories.
	Data utility: We expect that our open datasets will be useful to 1) researchers, mainly the drug spectra generated (to progress the state-of-the-art further), 2) industrial partners (to develop new technologies based on vapour detection), and 3) public authorities (to solidly implement the vapour technology for luggage screening).
Findability	Discoverability: Open datasets will be made findable with persistent links and a clear metadata schema.
	Persistent identifiers: We will use the ORCID reference (Open Researcher and Contributor ID) for researchers in our team. ORCIDs are valuable because they connect all of a researcher's output regardless of the author's name or affiliation changes. For digital research materials, we will use the

	Direct Object Identifier (DOI). It can be assigned to articles, or any other shared project output.
	Naming conventions: For public project documents, the naming convention will be based on the hierarchical structure of the project. The parent file will be the title of the work package, and each work package file will contain a minutes file as well as files for each task within the work package. The file will contain the meeting minutes for the work package, while the task files will contain documents about that task's activities and may potentially contain an additional minute's file containing meeting minutes specific to that task.
Access	Data repository: Gitlab to store, publish, and collaborate on open, machine-readable datasets Open data and access methods: Gitlab as the primary data repository, Zenodo for project data identifiers, Frictionless to describe, extract, validate, and transform tabular data.
Interoperability	Metadata vocabularies, standards and methodologies: We will employ various types of metadata to enhance the management and accessibility of our resources: • Descriptive: This type of metadata will aid users in discovering and identifying resources. It includes elements such as Title, Abstract, Author, and Keywords. • Administrative: Administrative metadata will be used to manage information resources efficiently. This includes information related to the resource's provenance, processing history, and rights management. • Structural: Structural metadata will describe the relationships between different parts of an information resource, particularly relevant for digital resources comprising multiple files. • Technical: This type of metadata will document the technical attributes of digital objects or resources, providing insights into their technical characteristics. • Preservation: Will be used to support and document the long-term digital preservation of information resources and digital objects, ensuring their continued accessibility and integrity
Reusability	Documentation: All our documentation will include basic information about the data that allows for its correct interpretation and reuse by ourselves in the future and other researchers (e.g., README file, README tab, Data Dictionary, Codebook, Commented Code, Lab Notebooks, as applicable). Licences: Each dataset used in the project will be accompanied by a data licence that details the permissions associated with the use of that dataset. The licences for each dataset will be stored in their respective repository, where they will be found in the respective dataset's README and metadata file.
Security	GitLab provides Rake tasks for backing up and restoring GitLab instances. An application data backup creates an archive file that contains the database, all repositories and all attachments. GitLab SaaS maintains SOC 2 type 2 and SOC 3 Type 2 reports concerning security criteria. Noteworthy, only the public data will be included in the databases. The sensitive and classified information will be handled according to the security guidelines.
Ethics	No major ethics issues have been identified in the PENTION project, no personal data will be reported and for that reason, the GDPR is not a big topic to work out. However, during the testing, some items will be screening (cars, belonging, etc) and this will be done under the Horizon Ethics guidelines. The correct implementation of ethics rules will be monitored by STAM.

#@CON-MET-CM@# #@COM-PL-CP@#

2. IMPACT #@IMP-ACT-IA@#

The SSRI-01-02 topic under the Horizon funding program indeed aims to foster innovative solutions from SMEs to address security challenges in the EU. The PENTION technology aligns well with these goals, particularly in combating **drug trafficking**, which directly linked to the Fight Against Crime (FCT), a significant issue in the EU as it is widely described in section 1. Drug trafficking **is not only related with the fact of transport drugs**, it also includes the whole processes from the acquiring of the precursors to the final distribution of the drugs.

As it is commonly known, drugs have detrimental impacts for citizen, **not only for the consumers' health and economic**, some other relevant impacts have been identified after a deep analysis of international reports, most of them are interconnected and have clear dependences:

- **Increased Healthcare Costs:** Drug addiction places a significant burden on healthcare systems. The costs associated with treating drug-related illnesses, overdoses, and infections like HIV and hepatitis can strain healthcare resources and lead to higher medical expenses for everyone, and for the rest of the society,
- **Crime and Violence:** Drug addiction is often linked to criminal activities, including theft and violence. This can create unsafe communities, increase crime rates, and put additional pressure on law enforcement,
- **Family Disruption:** When a person battles drug addiction, their family members often experience emotional and financial stress. This can lead to family breakdowns, child welfare issues, and an increased need for social services which is associated to more violence and reduction of the community well-being,

- **Workforce Productivity:** Drug addiction can lead to decreased productivity and absenteeism in the workforce. This not only affects individual job performance but also has economic repercussions for businesses and the broader economy,
- **The strain on Social Services:** Substance abuse puts pressure on social services such as addiction treatment programs, emergency shelters, and welfare systems. Increased demand for these services can strain available resources and funding,
- **Community Well-Being:** Communities affected by high rates of drug addiction can experience a decline in overall well-being. This includes reduced property values, diminished quality of life, and a sense of insecurity.
- **Stigmatization:** People battling drug addiction often face social stigma and discrimination. This can hinder their ability to seek help, maintain relationships, and reintegrate into society,
- **Cycle of Addiction:** The societal impact of drug addiction can perpetuate a cycle, as children growing up in environments affected by addiction may be more vulnerable to substance abuse themselves.

In order to evaluate the impact of drug trafficking on the society, it has been identified a potential impact on 9 out of the 17 sustainable development goals defined by the UN (marked in green circles in Figure 20), which means more than 50%, so if we, as global planet, want to progress on a sustainable development, the fight against drug trafficking should be a priority.



Figure 20 – Impact of drug trafficking on the sustainable development goals defined by the UN

However, the data confirms the opposite: **Drug trafficking is growing year by year** as the INTERPOL Secretary General Jürgen Stock said at the INTERPOL’s 50th European Regional Conference in 2023: *“Over the last five years, [drug] trafficking and consumption have increased by an order of magnitude, with Europe one of the main transit and destination markets”*⁴³. This statement is devastating and demonstrate the **need for action!** The PENTION solution aims to mitigate this by providing advanced screening technologies for illicit substances, potentially reducing the identified impacts. This not only meets but exceeds the requirements of the SSRI-01-02 topic by offering a robust and efficient solution for enhancing security and public health.

2.1 Project’s pathways towards impact

Currently, only a small fraction of drugs and their precursors are detected, whether in clandestine laboratories where they are synthesized or at customs checkpoints and also at courier facilities. This detection process mainly relies on a **risk assessment model** but is hampered by the sheer volume of cargo being transported (containers, parcels, vehicles, etc) and the ease with which clandestine labs can be concealed in industrial or rural areas as it was identified



Figure 21 – Example of criminal investigation dismantling a clandestine lab for drug production (source: EFTM)

by the authorities in the challenges (Table 1), primarily due to the **limitations of existing inspection techniques**. Often, the results in criminal investigations to identify potential labs or manual screening of parcels at checkpoints, both of which are costly in terms of equipment and personnel, have varying effectiveness in detection, high FAR, and lengthy analysis times (hours or days).

To overcome these obstacles, a multi-layered approach is essential for the LEAs and customs administrations, with direct applications for security forces. By **integrating innovative technologies**, detection capabilities can be enhanced, reducing reliance on criminal investigations and physical inspections, which are both costly and time-consuming. This is why we propose an innovative,

cost-effective, highly accurate, and rapid inspection method based on **vapour detection** which will serve as a primary screening system and can be combined with other technologies to improve detection performance.

⁴³ <https://www.interpol.int/News-and-Events/News/2023/Europe-Drug-trafficking-organized-crime-increasing-by-an-order-of-magnitude>

2.1.1 - Contribution to the Expected Outcomes

The PENTION project will **address the five expected outcomes** (from the call text) and support the overall EC ambitions (call destination), see the analysis in Table 9.

Table 9 – PENTION project expected outcomes

Expected Outcome 1: Development of a mature technological solution addressing EU security policy priorities in the areas addressed by the Cluster 3 work programme.
PENTION will focus on the area of Fighting Crime and Terrorism (Option A), aiming to enhance the detection of drugs and drug precursors in both criminal investigations, local manufacturing and checkpoint scenarios. The project will develop a comprehensive and advanced screening technology that incorporates several key features: • Wide Range of Substance Detection: Capable of identifying and detecting drugs, drug precursors, degradation products, etc, which is essential for the focus of the PENTION project: synthetic drugs. The database can be as extensive as required. • Rapid Analysis: Approximately 3 minutes for sampling in open areas and 1 minute for small volumes (parcels) + 1 minute for the analysis • Non-Intrusive Screening: Able to detect substances from clandestine laboratories and their waste in a distance, as well as precursors in parcels at courier and postal service facilities, without unpacking or opening parcels, thus preserving cargo integrity and privacy. • High Detection Rate (DR): Greater than 80%, depending on the type of substance and sampling conditions. • Automatic Results: No need for human interpretation, reducing errors and potential external “interests”. • Portable and Deployable: Ready for timely deployment across various settings. • Additionally, the project will place significant emphasis on synthetic substances, including NPS, aligning with the EU’s priorities as outlined in the “European Anti-Fentanyl Drug Control Plan”⁴⁴ and the real needs of Limburg region, with a direct impact to the rest of the EU and the world.
Related Deliverables: • D1.1: PENTION Specifications • D2.2: Chemical characterization of synthetic drugs • D3.1: PENTION embarked system • D4.2: Results of PENTION testing
Related Milestones: • MS2: PENTION technology developed, integrated and validated in-house • MS3: PENTION technology successfully validated in different real-world environment (use cases)
Related KPIs: • KPI#1: Reducing sampling time • KPI#2: Reducing analysis time • KPI#3: Number of substances in the database > 6 • KPI#4: Duration of the batteries in the sampler > 5 hours • KPI#5: Detection limit = ppq range (versus ppb/ppt dogs). This is the key to detection at the bulk level. • KPI#6/7: DR (Detection Rate) > 80% for open areas / > 90% for semi-close areas • KPI#8: FAR (False Alarm Rate) < 5% • KPI#9: Automatic Results (Level of human interpretation) = 0%
Expected Outcome 2: Facilitated access to the civil security market for small innovators.
Despite the undeniable innovation capacity of EU SMEs, they often face challenges in accessing public markets. These challenges include bureaucratic hurdles in public contracts, difficulties in reaching new customers, securing finance, industrial competition, and IP valorisation. These issues are often more pronounced in restricted markets, such as security. PENTION aims to overcome these barriers by assembling a diverse consortium of project partners. This includes four well-established SMEs specializing in cutting-edge security technologies, 2 of them from the EU and supported by another 2 with vast international experience. Additionally, the consortium features a LEA and a EU University. The PENTION proposal includes a solid support from Associated partners, including one LEA, one Customs authority, and a region regional organization including police, customs, etc, which will act as catalysts to facilitate SME access to the security market by promoting and supporting the PENTION prototype system. Additionally, support will be received from the AB which currently includes a security expert group (KEMEA), one big industry and a LEA. Furthermore, the involvement of a big industry like Smiths Detection (AB) with extensive Pre-Commercial Procurement experience and with many solutions in the international market will ensure greater exposure of the PENTION solution to all relevant stakeholders.

⁴⁴ <https://politique.pappers.fr/question/european-anti-fentanyl-drug-control-plan-QECR963108>

Related Deliverables: • D5.2/D5.3: Communication and Dissemination reports • D5.4: Stakeholder engagement and consultation strategy guidance • D5.5: Market introduction strategy, business plan and stakeholder engagement
Related Milestones: • MS4: Market introduction strategy defined and developed
Related KPIs: • KPI#11: Prototype TRL = 7 at the end of the project • KPI#13: Screening capacity for open areas > 100 m²/hour • KPI#15: Attendance to 4 customs/security related events • KPI#16: Communication: 150 media practitioners following the project results • KPI#17: Collaboration: 4 seminars organised with related EC projects
Expected Outcome 3: Improved cooperation between public buyers and small supply market actors for a swifter uptake of innovation in response to short to mid-term needs.
The PENTION proposal seeks to enhance collaboration between SMEs and security end-users, aiming to maximize the potential for rapid adoption of the proposed technological advancements. The consortium includes SMEs from Spain and Italy within the EU, and from the UK and Canada outside the EU. Additionally, the involvement of the Netherlands Police and the Limburg Region (associated partner) establishes a strong foundation for improved cooperation between SMEs and security end-users, at least in the Netherlands and Spain, which will be the first to adopt the PENTION solution. Additionally, the exploitation plan will include the distribution management plan which has the objective of working in collaboration with market experts to reach the final “public buyers” which are typically public administrations and their acquisition processes are based on complex tender procedures.
Related Deliverables: • D5.4: Stakeholder engagement and consultation strategy guidance • D5.5: Market introduction strategy, business plan and stakeholder engagement
Related Milestones: • MS4: Market introduction strategy defined and developed
Related KPIs: • KPI#15: Attendance to 4 customs/security related events • KPI#16: Communication: 150 media practitioners following the project results • KPI#17: Collaboration: 4 seminars organised with related EC projects
Expected Outcome 4: Stronger partnerships between small and medium EU security industry and technology actors to ensure the sustainability of the EU innovation capacity in the civil security domain and reduce technological dependencies from non-EU suppliers in critical security areas.
The PENTION proposal aims to foster versatile collaboration between SMEs and a research/technology institution to facilitate the transfer of innovative technologies, thereby enhancing the innovation capacity of EU SMEs and start-ups in civil security. The interest of Smiths Detection, main security player worldwide, is key for the partnership between the innovation developed by the EU SMEs and the big Industry. Additionally, the involvement of the Limburg Region in the PENTION consortium supports this goal in two key ways. Firstly, it amplifies the role of the Netherlands Police in the project through their long-term collaboration. Secondly, it creates direct communication channels between PENTION SMEs and other LEAs and relevant stakeholders in the Netherlands, such as Port Authorities, Airport Authorities, and various cargo logistic operators. These entities, involved in criminal investigations and security checks for parcels, could be interested in adopting the PENTION system.
Related Deliverables: • D5.4: Stakeholder engagement and consultation strategy guidance • D5.5: Market introduction strategy, business plan and stakeholder engagement
Related Milestones: • MS2: PENTION technology developed, integrated and validated in-house • MS3: PENTION technology successfully validated in different real-world environment (use cases)
Related KPIs: • KPI#10: Days of Real Testing in an Operational Environment (> 2 weeks) • KPI#12: Number of scanned items in the project (> 35) • KPI#14: Dissemination: 3k audience reached by our mainstream articles and press releases • KPI#15: Attendance to 4 customs/security related events
Expected Outcome 5: The SSRI Destination will offer an open environment to create knowledge and value through research in matters (including technology, but also social sciences and humanities) that are not exclusive to only one security area, but cross-cutting to the whole Cluster.

The proposed PENTION technology will significantly exceed the specific objectives of this call, addressing most of the security domains outlined in the Horizon Europe Cluster 3 Work Programme. Under the broad umbrella of Civil Security for Society, the PENTION project aims to impact **three of the four traditional security areas** within the cluster, while being developed under the **FCT framework**:

- a. **Fight Crime and Terrorism:** The primary operational focus of the PENTION technology is to enhance detection capabilities for drugs, precursors, and other illegal substances. This is crucial for combating organized crime and reducing terrorism risks across the EU.
- b. **Border Management:** Targeted mainly at customs agencies and law enforcement authorities, the PENTION technology can significantly improve border management by quickly and effectively scanning large volumes of goods, benefiting both national and EU levels.
- c. **Disaster-Resilient Society for Europe:** By detecting chemical and biological threats, the PENTION solution can substantially contribute to the resilience of European societies against CBRN (Chemical, Biological, Radiological, and Nuclear) emergencies and attacks.

Related Milestones:

- MS2: PENTION technology developed, integrated and validated in-house
- MS3: PENTION technology successfully validated in different real-world environment (use cases)

Related KPIs:

- KPI#14: Dissemination: 3k of audience reached by our mainstream articles and press releases
- KPI#15: Attendance to 4 customs/security related events
- KPI#16: Communication: 150 of media practitioners following the project results
- KPI#17: Collaboration: 4 of seminars organised with related EC projects
- Moreover, **all KPIs of PENTION proposal are related to the different areas of the Cluster**

2.1.2 - Contribution towards wider impact

Additionally, PENTION contributes to the achievement of the impacts outlined in the **Horizon Europe Strategic Plan 2021-2024**⁴⁵, fostering innovation and addressing societal challenges.

“Crime and terrorism are more effectively tackled [...] thanks to more powerful prevention, preparedness and response, a better understanding of related human, societal and technological aspects, and the development of cutting-edge capabilities for police authorities and infrastructure operators.”

The Strategic Plan expects R&I to bring improved prevention, investigation and mitigation of impacts of criminal acts. R&I should support police authorities in better tackling crime, as well as the different forms of serious and organised crime (such as smuggling, trafficking of illicit drugs) by developing new technologies, tools and systems. Besides, R&I actions are expected to develop operational tools for enhanced criminal investigation capabilities for police authorities. As such, PENTION will enable prevention of criminal acts (drug trafficking) by offering a rapid and efficient detection system with the capacity to be easily adapted to many use-cases and scenarios. With its enhanced detection capabilities, PENTION supports fight against criminal activities, avoiding disruption for civil society and delays for goods transportation. Moreover, PENTION will address the EU Drugs Strategy 2021-2025, aligning seamlessly with several of the strategic priorities outlined:⁴⁶

- 1) **Disrupt and dismantle high-risk drug-related organised crime groups operating in, originating in or targeting the EU Member States.** The priority area 1.2 states that the EU should “Track, trace, freeze and confiscate the proceeds of and instruments used by organised crime groups involved in the illicit drug markets”. Focus should be given to tracking, tracing, freezing and confiscating the huge criminal assets obtained from drug trafficking and related offences in order to remove organised crime groups’ capacity to engage in future crimes and to infiltrate the legal economy. Furthermore, measures to limit the criminal use of instruments that facilitate the illicit drug trade, such as materials used in drug laboratories should also be considered. PENTION clearly addresses this priority by developing technologies to detect clandestine laboratories where illicit drugs are produced, and detect precursors in the supply chain, limiting criminals’ capacity to produce drugs.
- 2) **Increase the detection of illicit wholesale trafficking of drugs and drug precursors at EU points of entry and exit.** The priority area 2.1 states that the EU should “Counter the smuggling of drugs and drug precursors in and out of the EU by using established legitimate trade channels”. Major known ports, airports and land entry and exit points in the EU that are used as hubs for the wholesale trafficking of drugs and drug precursors should be put on a high priority list for action targeting drug trafficking. PENTION meets this need by developing a detection technology suitable for detecting and identifying both drugs and precursors at EU border checkpoints.
- 3) **Tackle the exploitation of logistical and digital channels for medium- and small-volume illicit drug distribution and increase seizures of illicit substances smuggled through these channels in close**

⁴⁵ https://www.eeas.europa.eu/sites/default/files/horizon_europe_strategic_plan_2021-2024.pdf Page 58

⁴⁶ <https://www.consilium.europa.eu/media/49194/eu-drugs-strategy-booklet.pdf>

cooperation with the private sector. The priority area 3.2 states that the EU should “Target drugs trafficking via postal and express services”. Trafficking of drugs via postal and express services should be targeted, also paying attention to the impact of the COVID-19 pandemic. In this context stricter monitoring of shipments containing illicit substances is needed in close cooperation with postal and express services. PENTION addresses this need by developing a detection technology that can be used both for drugs and precursors at couriers and postal services facilities.

- 4) **Dismantle illicit drug production; prevent the diversion and trafficking of drug precursors for illicit drug production.** The priority areas 4.1 and 4.2 state that the EU should: “Counter illicit production of synthetic drugs” and “Tackle the diversion and trafficking of drug precursors and the development of alternative chemicals”. Further efforts of law enforcement authorities are needed to detect and dismantle illicit synthetic drug laboratories. Issues related to the diversion and trafficking of drug precursors and the development of alternative chemicals need to be addressed at European and international level. One of the key objectives of PENTION is to detect clandestine laboratories where illicit substances are produced, which is precisely reflected in the proposed use case 1 and which is a major challenge for Limburg region and Netherlands Police.

2.1.3 - Potential barriers

The following table includes the **identified barriers for PENTION implementation.**

Table 10 - Potential barriers and mitigation measures

Nature of the barrier	Description of the barrier	Mitigation measures to limit the impact of barriers
Commercial	To successfully commercialize the PENTION device, it must be adopted by “early adopters” such as LEAs and customs authorities. However, some potential users may be hesitant or reluctant to implement novel technologies , non-based on imaging.	The project includes numerous C&D activities (see WP5) to raise awareness among society, industry players, and users about the benefits of PENTION. Mitigation efforts will also involve real demonstration sessions to showcase PENTION’s superior detection capabilities compared to current methods. Additionally, the device will be user-friendly and safe to ensure widespread adoption by non-experts.
Commercial	Difficulty for SMEs to reach the market due to complex and long procurement and tender process	The commercial collaboration with big industry (like Smiths Detection) and local/national distributors is essential to overcome this barrier.
Technical	Part of the reluctance to implement innovative technologies can be due to two reasons: i) potential security vulnerabilities to cyberattacks and ii) integration challenges with legacy systems	PENTION concept will be designed for an easy integration with legacy systems in terms of data standardization. On the other hand, PENTION system will assure the protection against cyberattacks by implementing advanced cybersecurity solutions.
Logistical	Cargo, goods and parcel logistic flows are quite optimized and the implementation of new solutions requires modifications on these flows which can reduce the efficiency or even the need for additional steps in the flow chain.	One of the objectives of PENTION, mainly the PENTION-S version, is the reduction of the sampling time (KPI#1) and the analysis time (KPI#2) with the final goal of generating the minimal impact on the logistic flows.
Economic	While PENTION EC funding will demonstrate the technology’s high potential, additional funding will be needed post-project to extend demonstrations, increase the TRL, and finalize the PENTION product.	During the project, new funding alternatives, including public and private sources, will be explored. Potential agreements with major security players will also be considered.

2.2 Measures to maximise impact – Dissemination, exploitation and communication

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The consortium is committed to ensuring widespread recognition and seamless integration of the project into the practices of LEAs and authorities for detecting clandestine labs, drugs, and precursors. This section details several measures aimed at effectively **communicating and disseminating** the project’s progress and results, as well as outlining subsequent **exploitation** activities for the PENTION solution.

STAM, as the leader of WP5, will spearhead the development of a comprehensive dissemination and communication plan by M3. This plan will actively involve all consortium members, encouraging their input and contributions. It will remain dynamic, with regular updates to continuously refine and enhance the overall strategy.

2.2.1 - Exploitation activities and IPR management plan

The primary outcome of the project will be the innovative PENTION screener, developed in **two complementary setups**, both based on the same instrument:

- **PENTION-M (Mobile)** which will be mounted on a vehicle to detect clandestine drug production labs in the field. This version includes a portable sampler for easy and fast operation.
- **PENTION-S (Static)** which is for accurately and rapidly detecting drugs and precursors in static scenarios such as cargo, parcels, even luggage, etc. In this version, the sampling process is semi-static with, one static sampler for the normal courier flows and a portable unit for screening specific parcels/pallets.

The commercialization of the PENTION screener will target **three markets** according with the plan in Figure 22:

- **Market #1: PENTION – S.** Target stakeholders: **large courier companies** (e.g., DHL, FedEx, etc) and postal services, which could benefit from the static PENTION system installed at their facilities to detect precursors and chemicals in parcels. Entry into Market #1 will be earlier than the case of Market #2 and #3 since the product development program for the PENTION-S will be around one year shorter than the case of PENTION-M. It is expected additional work, including additional certifications, for the implementing PENTION technology in a mobile unit. Market #1 entry is expected for end 2029, two years after the completion of the project.
- **Market #2: PENTION - M.** Target stakeholders: **LEAs and local authorities across the EU combating drug production.** Following its launch in the EU market, global expansion will be pursued, leveraging the product's established presence in the EU. Market introduction is anticipated in 2030.
- **Market #3: PENTION – M.** Target stakeholders: **LEAs and customs agencies responsible for border control at customs checkpoints** (e.g., airports, ports, large logistics hubs). The static PENTION system will be marketed here, following a similar approach to Market #2. However, some customs agencies can be also interest in the static version (PENTION – S / Market #1).

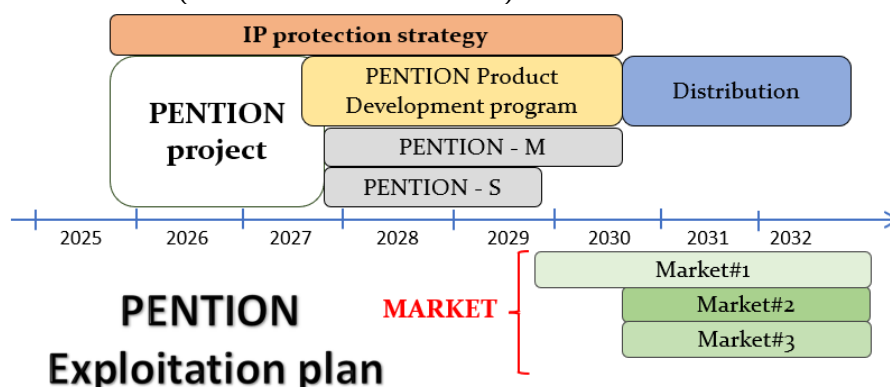


Figure 22 - Exploitation plan for PENTION technology

The exploitation phase will be coordinated by PPC, but MION will be the main exploiter of the technology, following an IP agreement with the other technology partners, mainly STAM. The business plan will cover the 2- to 5-year period beyond the project's conclusion, **targeting the three identified markets**. To maximize its impact, the consortium will also leverage the Horizon Results Booster to enhance its go-to-market strategy. A more detailed exploitation strategy will be developed during the project (WP5), alongside the planned technological advancements. During the last months of the project, the product development program will start and it will have an estimated duration of 2 years for the PENTION-S version and 3 years for the PENTION-M version. **Smiths Detection (AB) has shown its interest on supporting PENTION concept** which has a huge value since Smiths is the main and bigger security player worldwide with thousands of products released in the market during the last century, but no agreements have been done yet. The role of Smiths Detection is important to overcome some of the identified barriers in Table 10. However, the exploitation plan will also include other alternatives such as the **direct exploitation by MION**, including the product development, the manufacturing and the commercialization.

The exploitation plan will include a strategy to integrate PENTION technology within the [Customs Control Equipment Instrument](#) (CCEI), launched by the EC in 2021. This initiative aims to fund EU customs organizations to "maintain and upgrade state-of-the-art customs equipment such as new scanners, automated number plate detection systems, teams of sniffer dogs, and mobile laboratories for sample analysis.". Considering that the CCEI has incorporated the possibility of financing innovative equipment under the work program (recent update from the latest Multi Annual Work Programme), PENTION's strategy will focus on establishing a collaborative framework among Customs Authorities, especially with **Spanish Customs (Associated partner)**, potential stakeholders, and DG TAXUD. One of the objectives of involving Spanish Customs in the project, is the support to have access to the

CCEI program. This collaboration aims to validate PENTION technologies, creating an optimal environment to maximize the feasibility of their market transition. It is true that the CCEI program will conclude in 2027, just when the project is finishing, however it is expected that similar alternatives will be available from that time to continue supporting the update of screening technologies for customs. Therefore, the PENTION strategy will focus on engaging with potential stakeholders (customs organizations) to facilitate the **acquisition of some units during the program's final year** and to stay informed about potential extensions and new funding opportunities for similar customs programs. At the proposal preparation stage, the consortium has already developed a **preliminary exploitation plan** aimed at maximizing the PENTION project's impacts. This early strategy identifies potentially exploitable results and outlines the necessary steps for their exploitation, see Table 11.

Table 11 – Exploitable results in PENTION

Exploitable project result	Exploitation plan	Dissemination plan
PENTION-M	Commercial product	Organization of regular and thematic webinars, hands-on and collaborative workshops. Participation to industry trade fairs, scientific conferences. Cooperation with other innovative projects/solutions through knowledge sharing and joint initiatives. Social media campaigns, newsletters, press releases, website updates, public demonstrations.
PENTION-S	Commercial product	
PENTION-Sampler	Commercial product	

From the exploitation point of view, **three solutions will be developed in the project that will be a commercial product** after the product development program, the two versions of the PENTION screener (M:Mobile and S:Static), but also the portable sampler unit. MION has already developed a portable sampler unit (<https://vpscan-evd.com/>), however, this version requires the electrical connection to the power supply. A new product to be released from the project is an autonomous sampler based on batteries for the proper operation in the field. This PENTION sampler will be developed by STAM and a TRL7 prototype will be ready at the end of the project. Its exploitation is linked to the exploitation of PENTION-M version.

Within the exploitation plan, MION will develop the **distribution management plan**. Working in close collaboration with national distributors is key to have access to tender process and reach public administrations overcoming the commercial risks identified in Table 10. At the proposal stage, MION is in conversations with some EU distributors that have some interest on MION's vapour detection solutions, including the PENTION systems, the most advanced conversations are with [TARGET TECNOLOGÍA](#) for Spain region and [MICROSEGUR](#) for the Portugal region.

As discussed, PENTION technology is very versatile with more applications and markets, for example the aviation market, that the identified in the proposal, this is why a **deeper business plan** will be developed during the project and presented at M24.

Regarding **intellectual property rights (IPR) management**, PENTION aims to set a new standard for fast and reliable screening of drugs, precursors, and other illicit substances. An extensive search for similar solutions revealed **no existing patents covering the complete process (vapour collection + vapour detection)**, except some preliminary ideas patented in the nineties about detection of cocaine in large containers⁴⁷. ITI will support MION and STAM in the identification of the FTO (Freedom To Operate) and find the way to protect the technology. On the other hand, there are many patented solutions for parcel screening using traditional technologies like X-rays and for optimal item manipulation, none encompass the full PENTION concept. A solid IP protection strategy will be implemented (T1.3) by MION in collaboration with the PENTION partners, specifically with STAM. The main objective of this strategy is the protection of the technology; however, times are longer than the project duration and the goal is to complete the protection of the IP before the product enters in the market (see Figure 22). MION, with the support of the consortium, the IP protection firms and the patent attorneys and based on the Consortium Agreements (CA) rules, will decide whether to apply for new patents for the innovations developed. This step will safeguard the consortium's innovative contributions and ensure optimal utilization of project results. However, it is important to note that the budget for "IP protection fee/consultancy" will cover only the Freedom to Operate analysis, with potential patent filings occurring outside the project's scope and budget.

For PENTION project, **it is not planned that ITI (Canada) generates IP for the project, only supporting development activities according to its experience**. In the case that a conflict or disagreement occurs with non-EU partners, mainly with ITI, it has been already agreed that the applicable regulations are the Belgium-EU rules.

2.2.2 - Communication and Dissemination Activities

The consortium will implement an active communication and dissemination strategy to enhance the visibility of the PENTION solution among its potential users, including LEAs, authorities, customs authorities, security organizations, and logistic operators.

⁴⁷ <https://patents.google.com/patent/WO1998017999A1/en>

Important: some information generated in the project will be sensitive, so the SAB (Security Advisory Board) will assess the Communication and Dissemination (C&D) plan to be presented in M3. Some of the information generated in the project is sensitive, one important activity during the C&D plan preparation is the identification of the target audiences and define which type of information is focused for each audience.

Consortium **partners are already members of several industry associations and networks**, such as:

- **Security Organizations:** [EOS](#) (European Organization for Security), [AIAD](#) (Italian Federation of Aerospace, Defence, and Security Industries), [IAI](#) (International Affairs Institute), [CBNR-P3](#) (Italian Cluster on CBRNe), [START 4.0](#) (Competence Centre on the Security of Strategic Infrastructures), and [ICT-SMT](#) (Business Network on ICT, Security, and Territory Monitoring).
- **R&D Organizations:** [CERIS](#) (DG HOME's Community for European Research and Innovation for Security), [SIIT](#) (Ligurian Technological District on Integrated Intelligent Systems), [SOSIA](#) (Ligurian R&D Network on System of Systems and Intelligent Automation), and [EASS](#) (Ligurian R&D Network on Energy, Environment, and Sustainable Development).
- **Detection working groups:** [Detection Hub](#) managed by KEMEA
- **European Ports Alliance** is a Public-Private Partnership to step up the **fight against drug trafficking** and organised crime. Even though PENTION concept is more focused on the FCT topic, the concept is highly related to the control of goods and it is aligned with this EU initiative.

These associations and networks will significantly support the dissemination activities proposed for PENTION. The plan for the project is to be involved in additional working groups more focused on drug trafficking to enhance the potential of PENTION solutions, some relevant groups/organizations have been already identified: EUROPOL within the [Industry and Research days](#) and [EUDA \(European Union Drugs Agency\)](#).

For the communication and dissemination (C&D), the LEAs and customs authorities within the **associated partners and Advisory Board** will have also a relevant role. It has been included a specific task (T1.5) for the Advisory Board (AB) management since the objective of the PENTION project has shown a relevant interest in the community, however, during the proposal preparation, no resources have been dedicated to the management of this board and only few members have been included. The aim of the T1.5 is the management of the AB with presence of the most relevant partners, with international presence, and two main objectives: i) the definition of the requirements according with the real needs based on the initial use-cases defined, and ii) proper and efficient dissemination of the project results with the aim of preparing the way to the exploitation and final commercialization.

As far as **internal communication** is concerned, project information will be shared within the consortium through the established internal procedures. MION, as the exploitation partner, will also conduct competitive intelligence activities to keep partners informed about competitors, new market developments, and other relevant updates.

Communication of the results will be also done in **collaboration with other innovation projects** in similar fields, as the "Detection Hub" network is doing. Some synergies with other project are planned (see Table 12), with special emphasis with the [SAUST](#) project, which is aim at developing a baggage vapour screener for illicit substances. STAM and MION, are also involved in SAUST, aiming to **collaborate and enhance PENTION's detection capabilities** once implemented. The following table summarises the C&D activities will be carried out during the project:

Table 12 - Communication and Dissemination (C&D) activities planned in the project and their expected impact

Target audience	Channels	Timing/Expected impact	Type
Public institutions related to criminal investigation, customs, (national and international) such as DG Home, UNODC, World Customs Organization, etc	These stakeholders will be informed about the project and its outcomes through two main channels: 1. Direct Social Media Dissemination: Utilizing platforms like LinkedIn and Twitter to share updates and engage with the audience. 2. Active Dissemination on Target Audience Channels: Reaching out through specialized channels to ensure the information reaches the relevant stakeholders.	Starting in M813, we will develop a stakeholder engagement plan. This initiative aims to introduce the PENTION solution to key institutions involved in combating drug trafficking. By engaging these stakeholders early, we will pave the way for a smoother market rollout later on.	Diss.
Technological and industrial communities	Engaging in specialized conferences as speakers or through poster presentations to share insights and progress.	By the end of the project, we aim to have participated in at least four major events, including the World Police Summit, GPEC (General Police Equipment Exhibition & Conference), Security &	Diss.

	Participation both in the scientific parts and in the exhibition tables.	Policing, Milipol Paris, WCO forum, and Anticipate London. These events will provide valuable opportunities to showcase the PENTION solution to key stakeholders.	
	Technical dissemination: brochures and newsletters will be published with a summary of the technical specifications of PENTION.	Throughout the project implementation, we will produce various technical dissemination materials related to each project outcome. These materials will also include updates on the results of events and conferences attended, as well as details on collaborations and milestones achieved.	Diss.
	Workshop - Innovations in Security Screening: Exploring PENTION Technology.	In M12, STAM will organize a workshop inviting partners and key players in the sector to present an overview of the PENTION project. This session will cover the project's objectives and showcase the innovative screening technology developed in PENTION	Diss.
Public organisations and associations, general public	Establishing a PENTION website to host downloadable information about the project's results, including public reports, deliverables, events, and articles.	M1-M24. Project progress will be shared bimonthly on the website and social media channels, with active participation from all partners. We expect an audience of 150 media practitioners.	Comm.
	Targeted Dissemination Efforts: - Science Programs and Publications: The dissemination leader (STAM) will target major science programs and publications to present project progress and results. - Mass Media Outreach: Utilizing mass media to inform the broader public about drug trafficking prevention methods and how to identify and report counterfeit drugs.	M1-M24. These efforts will raise awareness of the PENTION solution and its capabilities among the general public and relevant organizations. We aim to reach an audience of over 3,000 viewers/readers.	Comm.
	STAM will coordinate the cooperation with the SAUST project to identify synergies and potential exploitation routes.	Conversations with the SAUST project will begin in M1 to synchronize the projects for the max amount of time.	Comm.
	Seeking synergies with other EC funded projects, particularly those under "CL3-2023-FCT-01-02": A Harmonized European Forensics Approach on Drugs Analysis" topic, but also with the 2024-FCT topics and the "CL3-2024-BM-01-05": Detection and tracking of illegal and trafficked goods" to explore new solutions for drug seizure.	Starting in M3, STAM will initiate discussions with funded projects under "CL3-2023-FCT-01-02" to promote shared insights and advancements in drug analysis techniques. An analysis of the 2024 funded projects will be done during the project implementation	Comm.

By implementing these strategies, the consortium aims to maximize the visibility and impact of the PENTION project, ensuring that its innovative solutions reach and benefit the intended stakeholders. In addition to the joint C&D activities outlined, each consortium member will undertake individual efforts. These will include publishing project results on their marketing channels, such as websites and social media profiles, and presenting them at conferences and workshops. PPC, with support from MION, will also develop an exploitation plan to determine the best strategy for introducing PENTION into the market in its two version: PENTION-M and PENTION-S. #§COM-DIS-VIS-CDV§#

2.3 Summary

SPECIFIC NEEDS (Specific needs that triggered PENTION project)	
Equipment: PENTION will deliver a detection system in two version: a vehicle-mounted (PENTION-M) and a static (PENTION-S), both for highly effective detection of drugs and precursors, with a focus on synthetic drugs .	
High accuracy: In order to enhance security measures, there is a demand for innovative solutions capable of precisely detecting drugs and precursors , while maintaining a minimal False Alarm Rate (FAR).	
Fast and efficient screening methods: Methods deployed by the PENTION project include a screening time between 2-4 minutes (sampling + analysis) depending on the use case, with a high Detection Rate (DR) of over 80% in open scenarios and 90% for semi-close scenarios. The project also ensures a low FAR < 5% and fully automatic results with 0% human interpretation.	
Database: The PENTION project aims to have a comprehensive and realistic database , encompassing several substances, with a focus on synthetic drugs with relevant presence in the EU currently (amphetamine, MDMA and methamphetamine), but with a direct application to NPS such as the case of fentanyl. The database will play a crucial role in achieving a high detection rate, in the ppq range.	
EXPECTED RESULTS (Results generated by the end of the project)	
Analyser unit. By the end of the project, the consortium will have developed a TRL7 prototype of the analyser system (2 versions). This prototype will enable LEAs, authorities, and customs agencies worldwide to rapidly and reliably inspect different scenarios, from parcels to clandestine labs producing drugs.	
Portable sampling Unit. The consortium will fully develop and optimize the sampling technology to make it portable and operable in several scenarios . This will enhance the system's flexibility and usability .	
AI-Based Detection Routines. The implementation of AI-based techniques will significantly impact the technology. Automatic peak and interferent detection routines will replace the current semi-manual operations, simplifying the implementation of new detection routines , especially in complex and noisy background spectra. Additionally, the intelligent atmospheric dispersion models will accurately provide an estimate of the drug sources.	
Enhancing detection capabilities. The system will combine new, highly efficient vapour collection techniques with a detection technology that integrates MION's Differential Mobility Analyser (DMA) with a Mass Spectrometer (MS). This combination will enhance the system's detection capabilities.	
Wider drugs database. During development, updated spectra of drugs and precursors will be obtained to increase libraries. This will ensure the system remains effective against evolving and new threats, including the characterization of fentanyl , as example of the NPS with devastating effects.	
DISSEMINATION, EXPLOITATION AND COMMUNICATION MEASURES	
Communication. Public communication will use social media and a dedicated website with regular updates on project progress and articles. The amplification of the AB , with international presence, is an objective of the project.	
Dissemination. The dissemination plan targets 3 main groups: public institutions (LEAs, customs, etc), technological and industrial communities, and public organizations + general public. Activities include direct communication, engagement with related projects, trade fairs, conferences, website publications, brochures, and newsletters to raise awareness of PENTION's results.	
Exploitation. Led by PPC, the exploitation phase involves developing a business plan, utilizing the Horizon Results Booster, and exploring integration with the CCEI program with an initial exploitation plan divided into three markets for the two versions of the PENTION system .	
TARGET GROUPS (Users of the results of the project)	
Market #1: Large courier companies (e.g., DHL, FedEx) and postal services, which could benefit from the static PENTION system installed at their facilities to detect drugs and precursors in parcels . Market entry is planned from 2029, after the product development program ends for PENTION-S version.	
Market #2: LEAs and local authorities across the EU combating drug production . The PENTION-M system will be utilized. Market introduction is expected in 2030 when the commercial product becomes available.	
Market #3: LEAs and customs agencies responsible for border control at customs checkpoints (e.g., airports, ports, logistics hubs). The PENTION-S will be marketed, following a similar approach to Market #2.	
OUTCOMES (Changes expected to see)	
Commercialization. The PENTION system is set to launch from 2029 , targeting EU law enforcement agencies (LEAs), customs, and courier companies according to the business plan.	
Adoption of Vapour Analysis. We aim to promote the adoption of vapour analysis by Customs agencies and authorities as a fast-screening method by disseminating project results. This aligns with the EU's CCEI initiative, launched in 2021, which supports the implementation of advanced detection technologies.	
Increased Parcel Inspection Rates. Once adopted by Customs agencies and couriers across the EU, PENTION's rapid and accurate technology will significantly enhance the fight against illegal drug trafficking. Its quick screening times will allow for higher inspection rates without disrupting the flow of goods, all at a low cost .	

Enhanced Resilience to new threats. The PENTION screener's accuracy will be continuously improved through a cloud-based database containing data on several drugs and precursors, including NPS that may emerge over time.

IMPACTS

Public Health. PENTION technology plays a crucial role in **safeguarding public health** by tackling drug consumption problems, with a special focus on NPS. This innovative technology helps control the manufacturing of drugs, ensuring that harmful substances do not reach the public. By doing so, it significantly reduces the risk of addiction and overdose, which are major concerns in many communities. This will reduce healthcare costs, as drug addiction burdens healthcare systems, raising expenses for treating related illnesses, overdoses, and infections.

Security and Safety. PENTION significantly enhances security by efficiently detecting and preventing the transportation and production of drugs, precursors which market **is directly linked with crime and violence**.

Civil society. By effectively identifying and limiting drug use, **PENTION will mitigate several drug-related impacts**, including **family disruption** (emotional and financial stress, breakdowns, child welfare issues, and increased need for social services), reduced workforce productivity, Community well-being, and the strain on social services (pressure on addiction treatment, shelters, and welfare systems, straining resources and funding).

Direct Impact on 50% of the sustainable objectives identified by UN.

3. QUALITY AND EFFICIENCY OF THE IMPLEMENTATION # @QUA-

LIT-QL@# # @WRK-PLA-WP@#

3.1 Work plan and resources

The PENTION project lasts 24 months and it is distributed in 5 work packages (WP) aiming to achieve 4 milestones (MS). After the definition of specifications (WP1) and during the first year of the project, the PENTION system will be technically developed (WP2); it will later be integrated, and validated under a controlled environment (WP3). During the last six months of the project, the PENTION prototype will be tested and validated in real pilot tests in an operational environment (WP4), mainly in Limburg region achieving the final TRL7. Many communication and dissemination activities are planned, aimed at increasing awareness of the PENTION project among key stakeholders, and also the development of the exploitation plan (WP5). The detailed effort is described in the following table:

Table 3.1a: List of work packages

WP	WP title	Partner #	Leader	PMs	Start	End
WP1	Management and requirements	1	MION	36	1	24
WP2	PENTION Technology Development	1	MION	82	1	12
WP3	PENTION mobile lab Integration and in-house testing	2	STAM	46.5	11	21
WP4	Validation, testing and pilots	6	NP	34.1	13	24
WP5	Dissemination, communication and exploitation	4	PPC	38	1	24
Total PM: 236.6						

WP	WP name and leader	T#	Task	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
1	Management and requirements (MION)	1.1	Overall project management																								
		1.2	Requirements definition				⊗																				
		1.3	IPR management																								
		1.4	Monitoring milestone compliance, risk management and compliance with regulations and standards																								
		1.5	Advisory Board (AB) Management																								
2	PENTION Technology Development (MION)	2.1	DMA-MS adaptation and portability																								
		2.2	DMA-MS chemical analysis and chemical development																								
		2.3	PENTION sampling system																								
		2.4	Development of SW detection routines																								
3	PENTION mob-lab Integration and in-house testing (STAM)	3.1	PENTION mobile lab embarking and integration																								
		3.2	SW integration and optimization																								
		3.3	In-house testing																								
		3.4	Validation under controlled environment																								
4	Validation, testing and pilots (NP)	4.1	Testing and use-case management																								
		4.2	SW Optimisation under real conditions																								
		4.3	Testing under operational environment																								
		4.4	Manuals and operators' documentation																								
5	Dissemination, communication and exploitation (PPC)	5.1	Dissemination and Communication																								
		5.2	Exploitation plan and Market introduction strategy																								
		5.3	Engagement with stakeholders																								

Figure 23 - PENTION project Gantt chart (Green cells indicates Milestones deadlines)

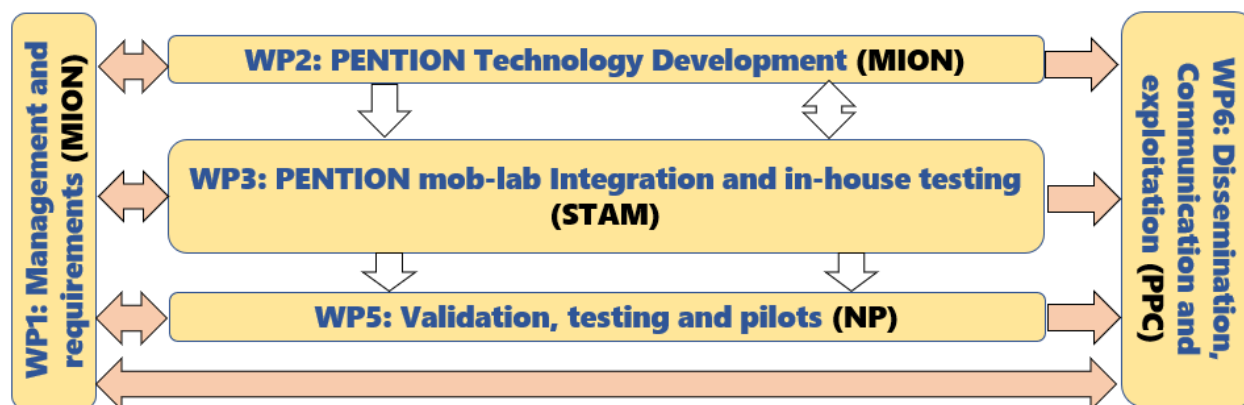


Figure 24 - Simplified PENTION project PERT diagram

Table 3.1b: Work package description

WP#	1
WP Title	Management and requirements
Objectives	
This WP will be implemented under the responsibility of the coordinator, MION, in order to support the project in succeeding its goals. Its main objectives are: • Ensure smooth and transparent implementation of the project, • Monitor project progress according to objectives and work plan in order to guarantee that related deliverables are produced on time, • Carry out administrative, technical, scientific, reporting and financial management, • Ensure efficient liaison with the EC and with related projects and initiatives, • Enable adequate coordination and communication between the work packages among partners, • Install mechanisms to effectively identify, assess and manage any potential ethical issue, as well as any potential administrative or technical risk, and develop mitigation strategies if needed, • Develop a comprehensive Data Management Plan (DMP), • Ensure data FAIRification and safeguard compliance of the project with the EC's Open Access policy, • Security Management related to Classified Information, • Ensure achieving the planned KPIs. Reassessment of KPIs along the development if necessary.	
Description of work	
Task 1.1 Overall project management [Lead: MION] (M1-M24, PMs:16) MION, as project coordinator, will ensure the smooth and transparent implementation of the project, maintaining an efficient relationship with the European Commission and projects related to border management. Coordinate and facilitate communication between the different work packages and consortium members includes the following activities: • Preparation of the project general management plan, • Maintain continuous communication with the European Commission and other relevant initiatives, • Coordination of the technical activities of the project, • Partner management, including associated partners (AP) and Advisory Board (AB) • The overall legal, contractual, ethical, financial and administrative management, • Overseeing the promotion of gender equality in the project, • Overseeing science and society issues related to the research activities conducted within the project, • Obtaining any financial security such as bank guarantees when requested by the Commission, • Chairing of project meetings in accordance with the management plan, • Quality management of the project documentation, including confidentiality management, • Monitor that the project is executed efficiently, within budget and on time, • Generate regular technical and financial progress reports, Task 1.2 Requirements definition [Lead: USN] (M1-M4, PMs:5) USN, with the support of all partners, including APs and the PENTION AB, will define the specifications and needs of PENTION technology; they will also update and validate the proposed use cases. Since vapour detection is a novel system and not widely deployed in the FCT environment, it is necessary to define the best approach to include it in the existing scenarios. USN will identify and specify the technical, operational and functional requirements of the project, in line with the needs of end users, such as LEAs and customs. One important requirement is the use of the PENTION concept after a positive signal proposing efficient cleaning procedures. A proposal for the validation tests (T4.3) will be also defined based on the proposed use cases. The final document	

will be reported in D1.1 at M4 reaching the first milestone (MS#1).

USN will coordinate the collection of information from different sources, but mainly from the interviews of police officers (NP and the AB) and experts in the new drugs sector beyond the AB, but also from the literature. A precise definition of the requirements is necessary to develop the PENTION solution (WP2) and achieve the KPIs. ITI, with the support of MION and STAM, will actively participate in this task supporting the definition of the requirements from the operational and technological perspective, thanks to their vast experience in new drugs detection and the implementation of screening solutions internationally.

Task 1.3 IPR management [Lead: MION] (M1-M24, PMs:6)

This task will ensure that all IP is identified as it is created. All identified intellectual property (IP) will undergo a comprehensive assessment by the PENTION consortium, with dedicated support from the technical partners (MION, STAM, and ITI). MION, being the primary technological developer, will play a key role in this process aiming to ensure appropriate protection measures for the IP before any public disclosure, taking into account the considerations and decisions of the owning party. Special emphasis will be put on designing appropriate technology transfer arrangements to facilitate the exploitation and commercialization of the technology as it will be described in the Exploitation plan and Market introduction strategy (T5.2).

From the project's start, MION, supported by the experience of ITI, will do a deep analysis of the patentability of technology trying to register a patent that covers the complete embarked unit as it is described in section 2.2.1. If the result of the analysis is positive, the application for a new patent will start at the end of the project after the consensus of the IPR terms with all the partners. The proposed budget for IP consultancy will not cover the new patent application, only consultancy activities to study the freedom to operate.

Task 1.4 Monitoring milestone compliance, risk management and compliance with regulations and standards [Lead: STAM] (M1-M24, PMs:5)

This task will have a twofold scope. First, to ensure timely identification, accurate assessment and effective management of any potential ethical issue. Even though no major ethics issues have been identified at the onset of the PENTION project, STAM will ensure that all project activities will abide fully by the EU General Data Protection Regulation (GDPR).

Second, this task will develop and update the Data Management Plan (DMP) considering Horizon Europe's standards. The DMP (D1.2 and the updated version in D1.3) will outline how collected, processed, and generated data will be handled during the project and after its completion, always pursuing data interoperability. It will outline: (a) how research data will be collected, processed or generated within the project; (b) what methodology and standards will be adopted; (c) whether and how this data will be shared and/or made open and/or can be easily found; (d) what legal, organisational and technical measures will be used to ensure its security and to safeguard the rights of individuals; (e) and how this data will be curated and preserved during and after the project. This task is designed to establish a robust data protection and privacy strategy that aligns with the EU's GDPR and other relevant legislation. Additionally, it aims to explore potential avenues for open data exploitation within the project. The task involves vigilant monitoring of data privacy and confidentiality considerations, ensuring strict adherence to legal and ethical standards throughout the data generation, utilization, storage, and sharing processes. The DMP will be developed by STAM, with support of MION, ITI and USN.

As the project progresses, it will be necessary to verify that all milestones are being met and that they are carried out in compliance with regulations and standards and taking into account the possible risks that may arise.

Task 1.5 Advisory Board (AB) Management [Lead: MION] (M1-M24, PMs:4)

During the proposal preparation some EU authorities, mainly LEAs, Customs administrations and Customs laboratories have shown the interest of the PENTION proposal because it covers an increasing and worrying challenge nowadays, however. At the time of the proposal submission, only Spanish Guardia Civil, Smiths Detection and KEMEA have confirmed their participation in the AB. In the case of project funding, MION will manage the PENTION AB with the objective to include more authorities, even from out of the EU. MION will encourage the AB to actively participate in the project both in the requirements definition and during the pilots.

WP#

2

WP Title

PENTION Technology Development

Objectives

The objectives of WP2 can be summarised as follows:

- To develop the PENTION detection technology up to TRL7,
- To adapt the DMA-MS technology to a mobile lab system in terms of portability,
- To develop the required AI-based software for vapour fingerprint analysis,
- To define and develop the operation procedures,
- To develop the detection routines for new drugs by vapour detection.

Description of work

Task 2.1 DMA-MS adaptation and portability [Lead: MION] (M1-M12, PMs:33)

Current DMA-MS MION's technology is developed and optimised for explosives cargo screening, however, the requirements for the case of drugs, and especially for synthetic drugs and NPS, are different. In addition to the chemical composition of the drugs, the proposed scenarios in PENTION are both open environments (e.g., clandestine labs) and semi-close environments (e.g., car screening, mail/courier screening). The present task aims at the adaptation of the DMA-MS technology to the new proposed scenarios and target substances in terms of portability and capacity of detection.

In terms of portability, current DMA-MS analyser has been developed as a fix unit to be operated in a lab/office environment, only the sampling unit (sampler) is portable, the aim of this task is to implement all the technical activities to provide portability to the analyser which includes:

- Integration of silent blocks in the structure of the analyser and the attachment points to minimize the vibrations of the instrument,
- Miniaturization and adaptation of the dimensions of the analyser which is a technical challenge,
- Reduction of the electrical consumption,
- Identification of critical elements and replace/adapt to be portable,
- Adaptation of the refrigeration of the analyser to the mobile lab.

MION will carry out the portability development in collaboration with STAM for the engineering issues.

In terms of detection, the collection filter and materials must be adapted to the vapour collection of the synthetic drugs. Current solution is optimized for explosive threats, and a new development is proposed in this task, including modifications in the absorption filter as well as in the thermal desorber. It is planned to test different new absorbents for the increase of selectivity which is key in the implementation of vapour detection technologies, and even more important for the case of parcel screening or open-air screening, simply because the variety of stuff and items available in the gas phase is greater. Some innovative active materials and polymers will be studied, adapted and validated under different conditions. Initially, a literature search is planned where some candidates will be identified and selected depending on their analyte affinity, dimensions, manufacturing capacity, temperature conditions, etc. ITI will do a selection of materials to be validated under different scenarios of potential interferents and target substances. In addition, a quantitative comparative analysis will be performed aiming at selecting the best candidate which will be extensively tested in WP3 and WP4 versus the current TENAX-based solution. Implementation of the new candidate will only proceed if its performance surpasses or equals that of the current TENAX (see Risk#2). ITI will coordinate this activity and MION will test and validate the proposed developments.

Task 2.2 DMA-MS chemical analysis and development [Lead: ITI] (M1-M12, PMs:26)

Synthetic drugs are chemically complex and a detailed chemical analysis is required for their chemical understanding and for the selection of the best strategies and markers for detection. In some cases, specific chemicals (precursors and the own drugs) require local synthesis lab. ITI has vast experience on this, e.g., some of the precursor's drugs are not commercially available, but can be custom made. In addition to the synthesis, one key activity in the project is the characterisation of the chemical compounds (degradation products), in order to perform a good understanding of these compounds, different and complementary analytical techniques are proposed, DMA-MS at MION's facilities and GC-GC-MS instruments in Canada (ITI). The combination of capabilities of MION and ITI is the perfect tandem for the proper chemical analysis. This task covers all the chemical characterizations of the target substances required during the first half of the project. ITI will propose and develop chemical strategies which will be validated by MION. It is important to remark that ITI will support with its experience the development of the chemical analysis which will have a clear benefit for the EU.

In this task, MION will also characterize the vapour pressure of fentanyl through the fentanyl standards in order to anticipate results to the mid-term scenarios in the EU and contribute to the consolidation of the scientific discussion described in the proposal.

Task 2.3 PENTION sampling system [Lead: STAM] (M1-M12, PMs:16)

The DMA-MS technology is quite versatile, mainly thanks to the separation of the sampling and the analysis processes which is essential for the purpose of the PENTION project. However, it is required an extra step in terms of sampling capacity and two samplers have been proposed for the PENTION project. For the PENTION-M version, the sampling requires some additional autonomy to have the capacity to screen complex areas in rural or industrial environments. The purpose of this task is the development of a wireless sampling system with no need to be connected to the electrical supply. ITI has vast experience developing portable and wireless system using high-efficient components. It is proposed to use COTS elements, mainly for the battery/charger units. This task

requires some development since the sampling requirements are quite demanding in terms of flow and pressure, and traditional blowers are out of reaching these requirements. This task will start with the selection of the elements and the performances in terms of autonomy, weight, portability, etc and two units are planned to be built.

Secondly, a fix sampling system is required for the PENTION-S version, STAM will develop and manufacture one unit for the implementation of the use-case#2. The main challenge for this version is the potential contamination after a positive detection. To prevent, avoid and control it, it is planned to use protecting films in the surfaces with direct contact with the vapours, additionally, some parts will be disposable after the use. These requirements will be defined in T1.2.

For the two versions, ITI will design and propose a concept, and STAM will manufacture the units, MION will test and validate them, and PPC will validate the developments from the end-users/commercialization perspective.

Task 2.4 Development of SW detection routines [Lead: USN] (M1-M12, PMs:7)

This task aims at overcoming the limitations of current semi-manual data processing, through the implementation of AI-based techniques that enable the identification of potential interferents (substance with similar molecular mass and ion mobility as the target substance).

USN will carry out a feasibility study on the most suitable techniques, and develop the most promising approach required for the implementation. Potential candidates are: multi-modal hybrid machine learning (e.g., OntoGPT⁴⁸), established control-theoretical modelling⁴⁹, Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS)⁵⁰. The selected techniques will be evaluated, trained and validated before the implementation using existing data from MION or additional datasets generated as per USN needs.

This task includes the implementation of atmospheric dispersion models by USN for the identification of the emission patterns and sources in the PENTION-M version with the objective of intelligent identification of the source of vapour emission and the final detection of the clandestine labs. To do this, some advanced atmospheric dispersion models will be developed to simulate the spread of vapours from clandestine synthetic drug labs, incorporating factors like wind speed, temperature, humidity, and terrain.

ITI will support the implementation of these models thanks to its experience on detection in open scenarios.

USN will develop two additional performances: i) the final operator interface, which is the main interconnection with the operator which will address the requirements identified in WP1 for user-friendliness, and ii) provides the PENTION system with the required cybersecurity solutions to avoid attacks in the PENTION analyser, STAM will support USN on this second activity.

WP#

3

WP Title

PENTION mobile lab integration and in-house testing

Objectives

The objectives of WP3 can be summarised as follows:

- To integrate the PENTION prototype in a vehicle in terms of hardware and operation,
- To validate the PENTION prototype in-house from the operational point of view,
- To validate the PENTION prototype under controlled environment for detection,
- To validate and demonstrate the detection of new drugs by vapour,
- To deliver a measurement system that is ready to use for the case of real scenario measurements.

Description of work

Task 3.1 PENTION mobile lab embarking and integration [Lead: STAM] (M11-M18, PMs:20)

The hardware prototype of the DMA-MS analyser optimised for portability in T2.2 will be integrated in a vehicle, most probably a van. Firstly, the most appropriate vehicle category will be identified, leaving some room for the integration of additional equipment, in case this is needed in the future development of the PENTION system. Supply systems for power and fluids will be selected to address the requirements of the DMA-MS identified in T1.2. The power unit may be equipped with a storage system, to ensure continuous operation of the sampling and analysis equipment and avoid fluctuations. A thermal study will be also implemented to define the optimal ventilation of the vehicle. The loading compartment will be reconstructed in 3D CAD to optimise the location of all necessary devices, and a mock-up of the integration will be constructed in laboratory, either at STAM or MION facilities. Afterwards, the vehicle will be procured, and the actual devices will be integrated. Finally, preliminary

⁴⁸ Caufield, J. H., Hegde, H., Emonet, V., Harris, N. L., Joachimiak, M. P., Matentzoglou, N., Kim, H., Moxon, S. A. T., Reese, J. T., Haendel, M. A., Robinson, P. N., & Mungall, C. J. (2024). OntoGPT (v1.0.4). Zenodo;

⁴⁹ (1982). Optimal control theory and economic analysis. In Gustav. Feichtinger (ed.), North-Holland Pub.

⁵⁰ Dubois, D., Grabisch, M., Modave, F. & Prade, H. (2001). Relating Decision with Uncertainty in Multi-criteria Decision Making Models. International Journal of Intelligent Systems (p./pp. 967-979), Wiley.

testing will be carried out on all the mobile lab equipment, to ensure functionality and pave the way for the validation campaigns. STAM will carry out the design and development, while MION will provide the requirements of the detector and ITI will supervise the integration in terms of detection capabilities.

Additionally, STAM will perform a safety assessment to evaluate the risks and regulations associated to mobile laboratories. The “homologation” process itself will take place out of the project scope.

Task 3.2 SW integration and optimization [Lead: USN] (M14-M21, PMs:4)

As soon as the hardware is integrated and operatively validated, the software routines developed in T2.4 will be integrated into the PENTION hardware by USN. The first optimization tests that will be carried out by the analysis of the first realistic data from real backgrounds and, if possible, positive signals coming from chemical standards in the lab, which will be served as the initial data to train a model by means of the target AI algorithms. A continuous monitoring of the algorithm progression by parameters included in the SW code will take place from USN team with the possibility to include modifications before the final validation. This activity will be executed in parallel with T3.3 (In-house testing) and T3.4 (Validation under controlled environment). To integrate this activity with PENTION targeted analytics capacities, USN will then integrate its own intelligence and high-quality data processing by means of a container-based approach and following state of the art Machine-Learning Operations (MLOps), guidelines and technologies (e.g., using Kubeflow or related tools).

The atmospheric dispersion models will be integrated in the PENTION prototype at this stage which will be also validated at lab level during this task by MION.

Task 3.3 In-house testing [Lead: STAM] (M16-M20, PMs:10.5)

Once the development of the two PENTION systems (hardware and software) are completed, in-house testing is required for the validation of the instrument in terms of functionalities, robustness and operability against the requirements defined in T1.2. This will include several types of testing such as (responsible partner):

- Mechanical validation of the system (STAM),
- Testing of operation after some trips and movements (STAM),
- Operation of the complete system for long periods of time (MION),
- Testing and optimization of the ventilation (STAM),
- Safety measures for proper operation (STAM),
- Software validation (USN),
- Capacity of detection (MION),
- Additional testing identified in T2.3/T2.4 (development) and T1.2 (specifications) (MION).

During this activity, a detailed analysis of the potential implementation risks will be performed, including the possibility to implement any of the mitigation (or correction) actions identified in Table3.1.e. Milestone #2 will also be reached during these tests.

Task 3.4 Validation under controlled environment [Lead: MION] (M17-M20, PMs:12)

The PENTION complete instrument will be ready for validation from M17; therefore, the objective of T3.4 is the validation of the detection routines from the detection point of view, according to the defined use-cases (see section 1) and other potential use cases defined in T1.2. MION, supported by ITI, will work on this validation.

The proposed scenarios will require the generation of simulated environments. To do this, commercial narcotic chemical standards will be acquired, samples from real seizures will be provided by the authorities (Netherlands Police and the AB) and legal substances that can be used as precursors. Other alternatives will be assessed before the final validation in the Limburg region (RIEC associated partner).

For the proposed validation by MION, we will follow an iterative process starting from the most difficult configurations (low-vapour concentration generated) using complex concealment processes to the easiest configurations where the vapour emissions are higher. The output of this task will be a quantitative analysis of the detection performances (Detection Rate) and a list of limitations found that can be solved before the final testing. One key parameter from this testing will be the optimal sampling conditions (aspirated volume and sampling rate). No testing of FAR will be performed at this point since not backgrounds will be analyzed.

In addition, the different parameters of the peak/interferent detection routines developed on WP2 and integrated into T3.2 by USN will be optimised in this task when real vapours from real substances are detected. However, this validation is only limited to detection since the main optimization will be done in WP4 where real scenarios will be implemented and the routines will be optimised for potential interferent identification.

WP#	4
WP Title	Validation, testing and pilots
Objectives	
The objectives of WP4 can be summarised as follows:	

- To manage the implementation of the use-cases in to the real-world environment at Limburg region,
- To install and validate the PENTION complete screener in an operational environment according to the defined use-cases and defined requirements,
- To validate the PENTION detection technology up to TRL7,
- To complete the manuals and other commercial documentation of PENTION system.

Description of work

Task 4.1 Testing and use-case management [Lead: NP] (M13-M22, PMs:4)

The PENTION pilots under realistic scenarios require effective management and active involvement from many partners under the coordination of NP and the main role of RIEC association (AP), specifically five additional teams covering the entire Limburg province, who conduct administrative checks on entrepreneurs in both rural and urban areas. This task will result in the detailed definition and scheduling of pilots in real-world environments, with opportunities to conduct pilot tests in areas where synthetic drug production is observed, expected, or predicted. Use case management will ensure ongoing updates to the pilot planning defined in D1.1.

This task also includes the management of all the requirements for the installation of the complete PENTION system in the operational environment by MION (coordinator of the technical installation) and STAM, but also with the support from NP and RIEC (coordinators of the installation from the logistics perspective) according to the requirements defined in D1.1. The installation includes both the physical installation of the screener, the tuning and the initial operational validation.

Task 4.2 SW Optimisation under real conditions [Lead: USN] (M20-M24, PMs:3)

The objective of task T4.3 is the optimisation of the AI-based routines by USN for the identification of interferences and target peaks in a controlled environment where some simulated scenarios were developed and tested according with the three defined use cases in section 1. However, the objective of this task (T4.2) is the optimization of the trained AI-based methods under conditions of real interferences where the variety of volatile compounds should be greater and more complex; specifically, an active learning exercise by means of the MLOps features developed and integrated in T3.2 shall be enacted to design self-adaptive and auto-optimising predictive models which are aware of contextual conditions represented by continuous monitoring of said conditions. To achieve this goal, numerous samples will be collected and analysed under various parameter configurations automatically. The study will focus on the hit rate in identifying anomalies in observed time-series data, with the aim of surpassing a hit rate that allows having a DR (Detection Rate) > 80%/90% (KPIs#6 and 7). The target DL algorithms will then be rigged to be continuously (re-)trained when more samples are analysed, so it is expected that the target FAR of <5% (KPI#8) will be reached at the project ends after extensive exercise of the active learning capacity.

Validation of the atmospheric prediction models under realistic environments in different days with different weather conditions and levels of concentrations will be also performed in this task.

Task 4.3 Testing under operational environment [Lead: MION] (M18-M24, PMs:24.1)

The objective of this task is to implement the PENTION project pilots according to the proposed scenarios defined in T4.1 and deliver the actual testing of the PENTION technology in an operational environment (TRL7) – KPI#11. MION will coordinate the testing and the work under this task will begin once the PENTION system is installed and the optimisation of the SW under real conditions (T4.2) is completed. Most of the testing will take place in the Limburg region (RIEC) supported by NP. STAM will actively participate in the installation of the system. The test plan defined in T4.1 will be executed with the active support of the facilities with the ultimate goal to achieve the Milestone #3 by the analysis of, at least, 6 substances (KPI#3) for around 2 weeks (KPI#10) and achieve the expected results in terms of robustness, DR (>80%/90% - KPI#6/7), FAR (<5% - KPI#8), with a detection routine with a sampling times of 1 minute / 3 minutes (KPI#1a and KPI#1b).

One demo day will be included in these testing in order to demonstrate the screening capacity of PENTION technology to stakeholders and potential clients, this demo day will be organized and managed by STAM.

For the pilots, the PENTION analyser will be used in all the pilots following the plan defined in the T4.1.

Task 4.4 Manuals and operators' documentation [Lead: PPC] (M21-M24, PMs:3)

This task consists of the elaboration of the manuals and commercial documentation. Within the manuals, the following documents are included: i) Installation manual, ii) Operation manual, iii) Maintenance manual, iv) Troubleshooting. The manual is reported in D4.3 (M24) and it will be mainly elaborated by PPC, which has vast experience in the development of commercial instruments for security. MION will provide all the information required for the documents. A training course, in presentation mode, will be also developed for this task; this material will also be used as a basis for production of commercial materials.

In this task, STAM will provide a technical study of the potential integration of the PENTION system in other mobile and portable platforms according to the requirements defined at the beginning of the project and also the

application for other purposes beyond the narcotics detection field.	
WP#	5
WP Title	Dissemination, Communication and Exploitation
Objectives	
The objectives of WP5 can be summarised as follows: • To ensure the visibility of the project and raise awareness on its activities and results among key user groups, • To strategically engage with related ongoing initiatives, aligning activities to amplify results, • To develop and implement a stakeholder engagement plan, ensuring project activities and outcomes respond to real needs through extensive consultation with a variety of stakeholders, • To ensure targeted dissemination and long-term impact and sustainability of project outcomes, • To develop the PENTION business plan and market introduction strategy.	
Description of work	
Task 5.1 Dissemination and Communication [Lead: STAM] (M1-M24, PMs:16) STAM will develop a plan for the dissemination and communication (D&C) of project activities and results. STAM will also set out the specific tools, channels and target audiences through which PENTION will share the results, updated according to partner and stakeholder feedback, and monitoring. The D&C plan will be first published by M3 (D5.1), then updated in M12 (D5.2), and M24 (D5.3). The D&C plan will be harmonised with the data management plan (DMP - D1.3) and will uphold the core focus on open-access data and publications. STAM will prepare the project's communication activities by developing a visual identity and associated templates, as well as establishing a public website and social media presence (Twitter and LinkedIn). To stay updated on the project advancements, all partners will engage in internal information-sharing mechanisms, promoting timely and impactful communication throughout the project duration. The main activities of the D&C plan are: • In M12, STAM will organise a workshop inviting partners and key players in the sector, to present an overview of the PENTION project as D&C activity. This session aims to cover its objectives and showcase the innovative screening technology designed for detecting illicit substances. Initially, this activity will take place online, but it will be assessed the possibility to do it in presential mode, • A demo is planned at the end of the project in Limburg region; STAM will organize and manage the demo to achieve the maximum diffusion and attendance to the event, • Cooperation with related working groups (Detection Hub, European Port Alliance, etc) and public organizations such as EUDA, EUROPOL, etc, • Cooperation with existing projects related to the PENTION topic, SAUST project from 2023 call and the selected projects in HORIZON-CL3-2023/2024-FCT and BM calls, • Cooperation and connection with the US "Opioid/Fentanyl Detection program"⁵¹. For this specific activity the role of PPC and ITI is essential, • Attendance to different commercial events in the field of new drugs (MION will coordinate this) Task 5.2 Exploitation plan and Market introduction strategy [Lead: PPC] (M13-M24, PMs:14) After the validation of the technology in the project, PPC, in collaboration with MION and STAM, will define the exploitation plan, the next steps to reach a final PENTION product and the future market introduction strategy. The exploitation plan will assess the possibility of reaching agreements with big security players, such as Smiths-Detection, to support the PENTION partners in reaching the final product (TRL9) and to define the market roadmap. As described, the PENTION technology is very versatile; the identification of new markets is planned in this task with the results of STAM in T4.4. All these plans will be developed during the PENTION project, and they will be reported in D5.5 (M24) and the Milestone #4 is expected to be reached at this moment. Task 5.3 Engagement with stakeholders [Lead: PPC] (M13-M24, PMs:8) In this task, PPC, with the support of all partners, will pursue active efforts towards engaging with key stakeholders. The idea is to promote the project with the aim of ensuring the usefulness of the solution for potential customers and end users. First, the team will search and document relevant stakeholders across all Europe in an internal database, in order to probe their specific needs and interests so that the project development will result in a solution that fits their requirements. This will include, but is not limited to, customs agencies, law enforcement agencies, and even policymakers at the national and international levels. Based on the aforementioned database, the team will develop a stakeholder engagement and consultation strategy	

⁵¹ <https://www.dhs.gov/science-and-technology/opioid-fentanyl-detection>

(D5.4, M18) to guide the consortium's interactions with them. The strategy will thus include guidelines for consortium partners on whom and how they should consult for the different project developments and how this consultation process should be carried out. The objective of this plan is to define a helpful strategy to gauge the needs and requirements of the target groups of the project results so that the consortium members can ensure these needs are met. This will mostly be carried out via a mix of surveys, workshops, and direct discussions.

Table 3.1c: List of Deliverables

D#	Deliverable name	Short description	WP	Leader	Type	Diss. level	Due
D1.1	PENTION Specifications	Specifications of PENTION technology and threats	1	USN	R	SEN	M4
D1.2	DMP	Data Management Plan Report	1	STAM	DMP	PU	M6
D1.3	DMP	Updated Data Management Plan Report	1	STAM	DMP	PU	M24
D2.1	PENTION adaptation and portability	Report of the design of DMA-MS for the PENTION system	2	MION	R	SEN	M12
D2.2	Chemical characterization of synthetic drugs	Report of the chemical characterization of PENTION target substances	2	ITI	R	SEN	M12
D2.3	PENTION portable sampler	Report of portable sampler developed	2	STAM	R	PU	M12
D2.4	PENTION SW detection routines	Report of the SW detection routines	2	USN	R	SEN	M12
D3.1	PENTION embarked system	Report of the embarked PENTION system and preliminary testing	3	STAM	DEM	SEN	M18
D3.2	Results of the integration of AI techniques	Report describing the achievements by implementing AI techniques in the detection routines	3	USN	R	SEN	M21
D4.1	Test and validation plan	Plan of the test campaign to be implemented in the PENTION project	4	NP	R	SEN	M17
D4.2	Results of PENTION testing	Report of the final results of PENTION validated in an operational environment, including NPS results.	4	MION	R	SEN	M24
D4.3	Manuals and documentation	Manuals and operator documentation	4	PPC	R	SEN	M24
D5.1	C&D plan	Communication and Dissemination Plan	5	STAM	R	PU	M3
D5.2	C&D report	M12-Communication and Dissemination updated report	5	STAM	R	PU	M12
D5.3	C&D report	M24-Communication and Dissemination updated report	5	STAM	R	PU	M24
D5.4	Stakeholder engagement and consultation strategy guidance	Guidance document on stakeholder engagement and consultation strategy to guide the consortium's interactions with stakeholders.	5	PPC	R	PU	M18
D5.5	Market introduction strategy, business plan and stakeholder engagement	Report of the market introduction strategy, the PENTION business plan, and the activities carried out to engage stakeholders	5	PPC	R	SEN	M24

Security: No classified information has been considered for PENTION project; however, a SAB, led by a PSO, has been already defined (see security Annex).

Table 3.1d: List of milestones

MS#	Milestone name	WP	Due	Means of verification
MS1	Requirements ready and validated by PENTION partners and AB	1	4	D1.1
MS2	PENTION technology developed, integrated and validated in-house	3	20	D3.1 & D3.2

MS3	PENTION technology successfully validated in different real-world environment (use cases)	4	24	D4.1 & D4.2
MS4	Market introduction strategy defined and developed	5	24	D4.3, D5.4 & D5.5

Table 3.1e: Critical risks for implementation

Type	Description and mitigation strategy	WP #	Prob	Severity
R1 - Technological	Vapour emission of synthetic drugs is not enough for detection	WP2-3-4	Low	Medium
The chemical composition of new synthetic drugs is not comparable to the composition of traditional drugs, for example the case of high pure cocaine, where the chemical composition is mainly based on the own molecule of cocaine. In this case, the composition of the volatile fingerprint is complex which is positive because there will be some high volatile molecules, but some key molecules can have low volatility and it can limit the detection in gas phase if the volatility is lower than the limit of detection (LOD) of the instrument. Therefore, it has been proposed the use of MION's DMA-MS technology for detection which is the most sensitive detection technology available in the market (0.01 ppq for plastic explosives – Ref [25]). If this happens and the detection in gas phase is not possible for one of the target substances, a deeper analytical analysis is required for the identification of new substances/markers with higher vapour pressure.				
R2 – Technological	Selectivity/Efficiency of the new collection materials is lower than expected	WP2-3	Low	Low
The objective of the new collection filter is to increase the selectivity of the collected vapours during the sampling, which means to have higher affinity to specific molecules. Some materials have been already selected from the theoretical point of view. However, it is possible that the final results are not as good as expected or other problems arise such as difficult cleaning, collection capacity reduction after many samplings, etc. It has been considered as an activity with low impact because current MION's filters are suitable anyway. The mitigation action implemented will be done in cooperation with the Risk#1 in terms of increasing the vapour concentration in the chamber if needed and use MION's filters.				
R3 – Technological	Cleaning process leading to false positives	WP2-3-4	Low	High
If the cleaning process is not accurate, target substances can be detected in the analyser from previous samplings. To mitigate the risk, longer cleaning processes cannot be accepted since the airport logistics does not allow interruptions. Thus, additional solutions are planned such as the use of disposable paper-based films or plastic sheeting to be removed after each sampling to reduce this risk and prevent contamination among samples. This mitigation actions have been already defined in T2.4 description.				
R4 – Technological	Dispersion models don't fit the real vapour behaviour	WP2-3-4	Low	Medium
Thanks for the latest progress on processing capacity and weather simulation models, atmospheric dispersion models have improved a lot recently, however, they depend on several factors and their accurate is not as good as expected. For the PENTION project, USN will implement the latest progress on these methods and validate that the behaviour of drug/precursor vapours follows the predicted flows, however it is possible that the accurate is not as good as expected. To minimize the impact of this risk (no detection of the vapours source), it is planned to carry out sensitivity analysis over available data, reducing scope or harnessing synthetic data generation to establish a proof-of-concept if and where needed.				
R5 - Implementation	Difficult access to real substances	WP3-4	Low	Medium
The access to target substances, mainly the three ones of interest for the project requires the active involvement of customs authorities and police forces, including associated partners and member of the AB. Some LEAs have already shown interest to provide samples, however it is possible that some difficulties are found during the implementation as it has happened in the past. The mitigation action will be the use of chemical standards and EMPK emitters. The capacity to generate vapours concentrations of these solutions can be similar to the case of one real sample in a difficult scenario. MION will also encourage all organizations from the project inception to avoid problems during the pilots.				
R6– Management/Financial	Lack of resources to meet all of the project's goals	WP1	Low	High
Given the vast complexity of the PENTION solution, there is a risk of miscalculating the needed resources. To mitigate this risk, we will develop a study of each task by each partner involved in them. Besides that, there will be extensive and continuous monitoring of the coordinator's technical development to avoid relevant deviations.				
R7 – Commercial/Policy	PENTION is not largely adopted by the target institutions	WP5	Low	Medium
If the relevant institutions, namely European Customs agencies and Police authorities, are not convinced by the capabilities of PENTION and do not decide to adopt it for their operations, the market rollout will fail or, at the very least, progress very slowly and only to a very small portion of the target market. In order to prevent this, an intensive dissemination activity will be done focused on the target market's features in order to make the solution known to potential customers and end users with the possibility of real demonstrations.				

Table 3.1f: Summary of staff effort (PMs)

The distribution of staff effort is shown in the table below, with more resources dedicated to WP2, where the technical developments will be carried out. However, the rest of the activities will require some relevant effort, from the management to the execution of the pilots and dissemination activities: these activities are considered very relevant for the validation and exploitation of the PENTION system.

#	Short	WP1	WP2	WP3	WP4	WP5	Total	%
1	MION	20	55	17	11	8	111	47%
2	STAM	10	8	23.5	11.5	16	69	29%
3	ITI	1	10	2	2.5	2	17	7%
4	PPC	2	3	2	3	9	19	8%
5	USN	2	6	2	2	2	14	6%
6	NP	1	0	0	4.1	1	6.1	3%
TOTAL		36	82	46.5	34.1	38	236.6	100%

Table 3.1g: ‘Subcontracting costs’ items

No subcontracting activities are planned for the PENTION project

Table 3.1h: ‘Purchase costs’ items

The following table summarizes partners with relevant “purchase costs” (>15% than personnel costs)

1/MION		
	Cost (€)	Justification
Travel and subsistence	18,000	Consortium meetings (4x1.500 = 6,000). Development, testing and validation travelling = 9,000. Dissemination activities = 3,000.
Other goods, works and services	107,000	MS maintenance (2 years) = 40,000; laboratory consumables and standards = 6,500; testing consumables = 5,500; mechanical designs and other materials for the filters = 5,500; electronics, thermal and hydraulic consumables = 4,000; dissemination costs (registration fees, material, etc) = 6,500; material shipment for testing = 5,500; IP protection fee/consultancy = 8,500; VAN = 25,000
Total	125,000	

2/STAM		
	Cost (€)	Justification
Travel and subsistence	18,000	Consortium meetings (4x1.500 = 6,000). Development, testing and validation travelling = 7,000. Dissemination activities = 5,000.
Other goods, works and services	48,000	20,000: Consumables and materials for the vehicle adaptation, sampling development, integration, etc. 28,000: Expenses for dissemination activities: 8,000 materials (brochures, posters, roll-ups, one-pagers), 2,000 website hosting and development, 8,000 fees and expenses for participation to conferences and fairs, 10,000 organization of workshops.
Total	66,000	

PENTION budget project justification

The following table represents the detailed budget distribution

Partner	Total Budget*	% budget	% PMs	Personnel	Travel	Materials	% materials/budget
MION	848,812€	41%	47%	554,050€	18,000€	107,000€	16%
STAM	555,625€	27%	29%	378,500€	18,000€	48,000€	11%
ITI	213,437€	10%	7%	148,750€	6,000€	16,000€	9%
PPC	212,062€	10%	8%	149,150€	12,000€	8,500€	5%
USN	126,875€	6%	6%	91,000€	10,500€	0€	0%
NP	91,312€	4%	3%	64,050€	9,000€	0€	0%

* Indirect cost included

The total budget of the project is 2.048 M€, of which 1.49 M€ is requested from the EC, as the 89% of the budget is allocated to SMEs (funding rate = 70%).

Even though most of the budget is spent on personnel (84%), a significant proportion is allocated to materials and travel. Travels (4%) are mainly for consortium meetings and pilots, but also for some dissemination events and conferences. Finally, around 11% of the budget is dedicated to materials, which is justified since the final objective of the project is the development of two new detectors. A detailed description of the budget distribution is below:

- **MION**, as project coordinator, contributes with the highest effort in the project (47% in personnel dedication (PMs) and 41% from the total budget which amounts 848 k€). MION is the main technology provider and **the proposed PENTION project is a real challenge** from the technical and scientific point of view which requires many experts working full time in the project and many consumables and materials to develop the proposed tools. Additionally, MION will provide a mass spectrometer to the project at no cost, but the maintenance service will be charged to the project since it will fully dedicate to this. Some consumables are required for the technical activities proposed in the project, mainly laboratory and testing consumables, new designs and developments, and all the costs associated to the consumables in the pilots. It is important mentions here that MION will acquire a van for the implementation of the PENTION-M version which is considered as consumable since it will be part of the prototype. MION, as main inventor and exploiter of the technology will pay for the costs related to fairs and events related with the future exploitation and commercialisation. Finally, some budget is dedicated to the protection of the technology. For the travels, MION will participate in the project meetings, but also in the validation of the technology in the pilots. MION has some budget to ship the PENTION analyser to the pilots, PENTION analyser is based on a MS which requires special care during the shipping, typically, High-tech courier companies are selected for this.
- **STAM** contributes with 29% of the personnel efforts and the 27% of the project budget. STAM budget is divided into 85% personnel and 15% of other direct costs (materials and travels). The main activity of STAM is the design of the PENTION portable sampler and the engineering adaptation of PENTION-M to the van; for these tasks of WP2 and WP3, several consumables are required, estimated in 20,000 €; these are costs for materials for the manufacturing of mechanical elements, mechanical actuators, fasteners, connectors, ventilation units, materials for cleaning, hydraulic systems, electronics and control, wiring, pneumatic actuators and sensors, temperature controllers and heaters, etc. STAM will validate the system at MION facilities and the final operational facilities; a 5,000 € budget is estimated for material shipping. Additionally, STAM will lead the dissemination activities with an estimated budget of 28,000 € which includes the generation of materials, website hosting, fees to participate in events and conferences and the organization of workshops.
- **ITI**'s main activity is the development of the chemical detection routines for the new threats in close collaboration with MION and the support in the chemical developments in terms of detection, the objective is to bring to the EU all the knowledge generated in Canada during the previous years which requires 10% of the budget which is divided into the 87% of the personnel resources and 16,000€ for chemical materials. Additionally, 6,000 € for travelling to the UE. It is estimated that ITI will physically attend the KOM, the final meeting and have an active participation during the pilots.
- **PPC**, leading PENTION project's exploitation activity, allocates 10% of the budget. Within this allocation, 88% is designated for high-skilled personnel responsible for implementing the exploitation plan, the business-related actions and the communication with stakeholders. PPC will have some budget for the attending of conferences in collaboration with STAM, from the exploitation perspective, mainly to engage potential stakeholders.
- **USN** requires the 6% of the budget which is broken-down into personnel with the 90% and traveling, since its main activity is the software generation. No material is planned to be acquired.
- Finally, the participation of **NP** is limited to the 4.5% of the budget which is divided into personnel with the 87% and traveling.
- **Associated partners** have not required any budget, they will participate in the project, mainly supporting the requirements definition and the pilots, at no cost. It is expected physical attendance to events and meetings, but all costs will be by their own.
- **Advisory Board (AB) partners** will not have any budget. The inclusion of **Smiths Detection (SMD)** in the AB has been done strategically, SMD is the major player in security internationally and it is quite interested on PENTION concept (synthetic and new drugs), so SMD wants to follow the developments in the project for a potential future exploitation of the technology and supporting a potential product development program.
- **Additional notes:**
 - No equipment is planned to be purchased within the PENTION project. The VAN proposed for PENTION-M is considered as material since it is part of the final PENTION-M prototype.
 - Budget for non-SMEs (NP) and RTO (USN) is lower than the 15% required in the call for proposals.
 - Associated partners and the AB will participate without budget.

3.2 Capacity of participants and consortium as a whole #@CON-SOR-CS@# #@PRJ-MGT-PM@#

The PENTION project aims to develop a reliable and accurate vapour screener for the new drugs detection and identification under a wide variety of use cases, but with a wider implementation in the short term, not only in the number of scenarios, but also the inclusion of new chemicals and substances. This will be done in a collaboration of **6 partners: MION, STAM, ITI, PPC, USN and NP**. This novel and advanced screening technology will be able to significantly increase the number of scenarios and cases that can be inspected by authorities (police and customs) and the screening efficiency, upholding privacy principles without significant impact on the courier flows and the privacy in the daily activities at home, in the case of the clandestine laboratories.

In addition to the partners, it is **highly remarkable the Associated Partners** (Limburg Region-RIEC, Spanish Tax office and Spanish National Police) and **the Advisory Board (AB)** (Guardia Civil, KEMEA and Smiths Detection). All of them have shown a great interest on the project concept since the uncontrolled proliferation of synthetic drugs is a big real problem in the EU. All of them will support the project at no cost.

This technological development, as well as the testing and validation of the resulting solution, requires a multidisciplinary approach that is well-addressed by the PENTION consortium (see Table 6 in section 1). The consortium is strongly committed to building a long-lasting and professional project management structure and a partnership network extending beyond the duration of the project itself, both for the preparation for commercialization of the PENTION screener and for potential collaborations in the future.

The consortium counts on **the vast experience and capabilities of each of its members under the coordination of MION**. Each member has defined roles and tasks, to integrate cutting-edge technology into the proposed solution to help end users, EU Customs agencies and later police officers, in their fight against the smuggling of drugs and other illegal goods, and also for explosive detection in terms of security. It is expected that the consortium's intrinsic value chain and its members' networks will be very helpful in ensuring not only outstanding project execution but also excellent subsequent commercial exploitation of PENTION throughout the EU and, eventually, globally.

Consortium partners include **different but complementary and balanced skills** (see short descriptions later on). They include experts in analytical instrumentation, instrumentation for air samples, high-advanced Artificial Intelligence development, multidisciplinary engineering, operation in “security” areas, pilot testing under different scenarios, dissemination and communication activities. This large range of skills working together, under strong and efficient coordination efforts, will ensure the successful completion of the planned development. In addition to the complementary skills, the PENTION consortium has an **EU cross-boarding approach** with the participation of 3 countries (Spain, Italy and The Netherlands), plus an important **international participation** with experts in synthetic drugs detection in Canada and in the introduction of innovative technologies in the security market in the UK.

The consortium has a **strong commitment to gender equality** and they will promote a gender balance both within individual organizations and at consortium level.

Short descriptions of each of the 6 consortium members, as well as the **4 associated partners**, are provided here:



Mobility Ion Technologies SL (MION) is a Spanish SME aiming at the development of analytical instrumentation based on its own analytical ion mobility technology which is the core of the PENTION technology. MION's technology is mainly focused on security applications for the detection and identification of illicit and dangerous substances in many scenarios. MION has relevant experience both in the development and validation of the technology in an operational environment, both in the aviation and maritime domain. MION will provide the basis for the screener technology that is central to the project and will actively lead the technological development, as well as participate in the testing and validation tasks in all the locations. In addition, MION will coordinate the PENTION consortium, being Daoíz Zamora (MION CEO) the project coordinator who has vast experience in EU funding programs throughout his professional career.



STAM S.r.l. (STAM) is an Italian multidisciplinary engineering SME which provides high-tech and turnkey solutions in Industry 4.0 & Robotics, Space & Defence, Security & Transport, Energy & Bio-circular economy. Since the company's establishment in 1997, we have been supporting our customers along all stages of the product design cycle. Since 2010 we have enlarged our expertise to sensors and measuring techniques as well as engineering and new designs to support our customers moving onto the emerging Industry 4.0 paradigm. The enthusiasm and the cross-sectoral experience of the personnel is our main asset jointly with our patent's portfolio and our 1000+ network of satisfied clients and partners across the EU and beyond.



INNOVATIVE TECHNOLOGIES Inc (ITI) is a Canadian SME aiming at supporting international companies in different fields, but with a high focus on chemical detection technologies providing innovative methodologies for air sampling in different scenarios like cargo, truck, cars, etc and for different purposes such as environmental monitoring, but highly focused on security and customs control. The company is headed by Dr. Sabatino Nacson with more than 30 years of experience and knowhow in new product development,

engineering, chemistry and managing.



Precision Plus Consulting Ltd (PPC) is a UK based SME focused on assisting innovators of ground breaking technology in the defence, security and counter narcotics markets. Led by Samantha Ollerton, an international recognised expert on the field of illicit material detection, PPC Ltd. has a proven track record of assisting companies in their product development and go to market strategies enabling business growth and improvements in the detection capabilities available to Government agencies and commercial enterprises.



The **University of Sannio (USN)**, specifically the Department of Engineering (DING) is a dynamic academic and research hub dedicated to advancing engineering sciences and technology. It offers a range of undergraduate, graduate, and doctoral programs focused on areas such as computer science, telecommunications, energy, civil engineering, and industrial automation. The department is renowned for its interdisciplinary approach, integrating theoretical knowledge with practical applications to address real-world challenges. In addition to its educational mission, DING is actively involved in cutting-edge research and development projects that span multiple disciplines facing real problems.



Netherlands Police (NP) plays a crucial role by offering operational expertise, outlining needs and requirements, and assessing results from a LEA perspective (end-user). NP officers contribute to the definition of realistic scenarios, coordination and active participation in the implementation of the use cases and the implementation of training programs for personnel and security professionals, the covering legal aspects of the proposed research and the dissemination of the project results.

ASSOCIATED PARTNERS: The **project is complemented by 3 relevant associated partners** which are:

- **RIEC Limburg** (Regional Information and Expertise Center Limburg) is a collaborative initiative in the Netherlands that brings together regional law enforcement, municipalities, and other public agencies to combat organized crime and complex criminal networks. Focusing on issues such as drug trafficking, money laundering, human trafficking, and fraud. RIEC Limburg facilitates the sharing of intelligence and resources among partners. By pooling expertise from different agencies, it enhances the capacity to address crime that crosses jurisdictional boundaries. RIEC Limburg emphasizes a multi-disciplinary approach, coordinating preventive, administrative, and criminal measures to disrupt illegal activities.
- **Spanish Tax Office.** The Customs and Excise Department is the competent Branch of the National Tax Administration (AEAT) Public Entity in Spain. As such, AEAT is the National Competent Authority with State delegation to implement the National Taxation System in all the territory of the State, National or Regional Administrations and other National Public Entities, as well as Customs duties and the financial resources collected on behalf of the EU. Customs duties include the fight against smuggling, counterfeit, illicit trade, money laundry and fraud among many other competences.
- **Spanish National Police.** The National Police is an armed institute of a civil nature, with a hierarchical structure whose mission is to protect the free exercise of rights and liberties and guarantee citizen security, with a scope of action throughout the national territory, differentiating itself from other Spanish security forces due to its clearly urban nature. Among its exclusive powers is the investigation and prosecution of drug-related crimes throughout the whole national territory.

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